SCRAPER

FULL-CIRCLE CASING SCRAPER

CLEANING

GENERAL DESCRIPTION

CASING SCRAPER THAT REMOVES MUD, CEMENT, BULLETS, RUST, SCALE, PARAFFIN AND PERFORATION BURRS FROM THE INSIDE WALL OF CASING. THE BLADES ARE SPACED TO CONTACT 600 DEGREES OF THE CASING SURFACE AT ONCE, SO EVERY PART OF THE WALL IS SCRAPED WHETHER THE TOOL IS ROTATED OR SPUDDED. SHORT AND COMPACT, WITH A LONG TAPER ON THE BLADES, IT PASSES THROUGH CASING JOINTS WITHOUT HANGING. IT WORKS IN VERTICAL OR ROTARY OPERATIONS AND CAN BE RUN ON DRILL PIPE, PIN UP OR PIN DOWN.

TECHNICAL CHARACTERISTICS

Scraper for casing OD	Casing weight (lb/ft)	Casing ID min.	Casing ID max.	Top connection API Reg.	Fishing neck Ø	Tool ID	Tool OD
4-1/2	9.5 - 13.5	3.886	4.622	2-3/8	3-1/8	13/16	3-5/8
5-1/2	14.0 - 23.0	4.662	5.250	2-7/8	3-3/4	1-1/4	4-3/8
5-1/2, dressed for 6-5/8	20.0 - 33.0	5.600	6.188	2-7/8	3-3/4	1-1/4	4-3/8
7	17.0 - 38.0	5.850	6.688	3-1/2	4-1/4	1-1/2	5-1/2
7, dressed for 7-5/8	24.0 - 47.1	6.308	7.188	3-1/2	4-1/4	1-1/2	5-1/2
9-5/8	32.3 - 64.9	8.224	9.188	4-1/2	5-1/2	2-1/4	8-1/8
10-3/4	32.75 - 79.2	9.22	10.412	6-5/8	7-3/4	3-1/2	9-3/16
13-3/8	48.0 - 98.0	11.911	13.238	6-5/8	7-3/4	3-1/2	11

Dimensions in inches. One blade size covers a full range of casing weights for each casing OD. Casing ID range = recommended minimum and maximum.



DESIGN FEATURES

- **Full-circle scraping:** two rows of blades overlap, covering 600 degrees of casing wall.
- **Spring-loaded blades:** move in or out with gradual ID changes and stay on the clean wall without cutting the metal.
- **Abrupt obstructions:** a burr or cement fragment is held by the close-fitting blade slot, the mandrel and the split retainer ring.
- **Wear-resistant blades:** precision-cast tool steel, tough, slightly work-hardening in service.
- **Secure construction:** springs fully enclosed, blades fit closely over the spring housing, so loss downhole is virtually impossible.
- **One-piece mandrel:** heat-treated alloy steel, pin by box, with a fishing neck at both ends.

APPLICATIONS

- Clean out cement, hardened mud and paraffin.
- Remove perforation burrs and embedded bullets.
- Dress burrs, nicks and tight spots left by bits, fishing tools or dents.
- Prepare a clean casing ID before running packers, patches, spears, swabs and wireline tools.

OPERATION AND MAINTENANCE

- **Run:** usually pin up, with a bit installed below; scrape by rotating or spudding.
- **After each run:** disassemble, clean, inspect and replace worn parts; blades and springs are removed after taking out the four retainer-ring screws.
- **Re-dressing:** a blade installation tool (required accessory) compresses the blades during assembly.

HOW TO ORDER

Please specify: 1. tool or part name 2. size and weight of casing 3. tool joint connection 4. recommended spares: one set of blades and one set of springs.

CIRCULATING SUB

BALL-DROP CIRCULATION SUB

DRILLING

GENERAL DESCRIPTION

SIMPLE, RUGGED CONVENTIONAL BALL-DROP CIRCULATION SUB IN A SINGLE-COLLAR CONFIGURATION WITH NO MID-BODY CONNECTION. IT IS EASY TO USE AND OFFERS VARIABLE CYCLE CAPABILITY. A NOVEL BALL DISPOSAL SYSTEM LETS THE BALLS BE EMPTIED WITHOUT DISASSEMBLY, ALLOWING FIELD SERVICE AND MULTIPLE RUNS. STANDARD API CONNECTIONS AND HIGH-STRENGTH MATERIAL OPTIONS FOR DEEP GAS ARE AVAILABLE.

APPLICATIONS

- Wellbore cleanout
- Managed pressure drilling
- Underbalanced drilling
- Cementing operations
- Workover and completions
- Stuck pipe mitigation operations

DESIGN FEATURES

- **Single collar design:** no mid-body connections.
- **Variable cycle capability.**
- **Standard API connections.**
- **Ball drop activation:** rugged and proven.
- **Ball disposal:** balls emptied without disassembly; field serviceable, multi-run.
- **High-strength material options** for deep gas.

ITEMS TO DISCUSS

- Customizable housing for increased cycles
- Operating temperature range for the application
- Repair and maintenance options for staged repairs
- Chemical compatibility with specialist drilling fluids
- Compatibility and operation with third-party downhole tools
- Installation, handling and operating guidelines

HOW TO ORDER

Please specify: 1. tool size and top and bottom connections (standard API) 2. number of cycles required 3. material grade (high strength for deep gas, if needed) 4. operating temperature range 5. drilling fluid and chemical compatibility 6. third-party tools in the string.

A high-temperature / high-pressure version is available for demanding applications: ask for details.

RCJB

REVERSE CIRCULATION JUNK BASKET

FISHING

GENERAL DESCRIPTION

JUNK BASKET THAT USES REVERSE CIRCULATION TO RECOVER SMALL JUNK THAT OTHER BASKETS WOULD MISS. FLUID IS JETTED OUT AND DOWN AGAINST THE FULL CIRCUMFERENCE OF THE HOLE, SWEEPS LOOSE OBJECTS TOWARD THE CENTRE AND RETURNS UP THROUGH THE BARREL AND OUT OF PORTS AT ITS UPPER END. A DRAIN THROUGH THE TOOL PREVENTS PULLING A WET STRING EVEN IF THE INNER BARREL IS PLUGGED BY CORE. IT RECOVERS BIT CONES AND BEARINGS, BROKEN SLIPS, WIRELINE, HAND TOOLS, MILLING CUTTINGS AND DEBRIS, AND CAN BE CONVERTED INTO A FISHING MAGNET.

TECHNICAL CHARACTERISTICS

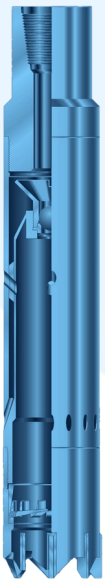
STANDARD TYPE (popular bit cone sizes)

Hole size	Barrel OD	Max. fish Ø	Top conn.	Teeth	Wt (lb)
3-3/4 - 4	3-5/8	2-1/4	2-3/8	6	85
4-1/8 - 4-1/2	4	2-1/2	2-3/8	6	98
4-5/8 - 5	4-1/2	3-1/16	2-7/8	8	108
5-1/8 - 5-1/2	4-7/8	3-9/16	2-7/8	8	128
5-5/8 - 6	5-1/8	3-3/4	3-1/2	8	142
6-1/8 - 6-1/2	5-3/4	4-3/8	3-1/2	8	185
6-5/8 - 7-3/8	6-1/4	4-3/4	4-1/2	8	238
7-1/2 - 8-1/4	7	5-1/8	4-1/2	8	293
8-3/8 - 9-1/2	7-7/8	6-1/16	5-1/2	8	374
9-5/8 - 10-5/8	9-1/8	7-1/16	6-5/8	10	441
10-3/4 - 11-5/8	10-1/8	7-5/8	6-5/8	10	576
11-3/4 - 12-1/2	11	8-5/16	6-5/8	12	665
12-5/8 - 13-5/8	11-7/8	8-1/2	6-5/8	12	797
13-3/4 - 16	13	9-13/16	6-5/8	14	932
16 - 17-1/2	15	11	6-5/8	16	1,563

W7R TYPE (W7R and other oversize cones)

Hole size	Barrel OD	Max. fish Ø	Top conn.	Teeth	Wt (lb)
3-3/4 - 4-1/8	3-5/8	2-3/4	2-3/8	6	78
4-1/4 - 4-1/2	4-1/8	3	2-3/8	6	85
4-5/8 - 5	4-1/2	3-3/8	2-3/8	8	95
5-1/8 - 5-1/2	4-7/8	3-5/8	2-7/8	8	108
5-5/8 - 6	5-3/8	4-1/16	2-7/8	8	141
6-1/8 - 7	5-7/8	4-5/8	3-1/2	8	173
7-1/8 - 7-1/2	6-3/4	5-1/4	3-1/2	8	223
7-5/8 - 8-1/2	7-5/16	5-5/8	4-1/2	8	280
8-5/8 - 9-1/2	8-3/16	6-1/2	4-1/2	10	380
9-5/8 - 10-3/8	9-1/8	7-1/8	5-1/2	10	407
10-1/2 - 11-5/8	9-7/8	7-3/4	5-1/2	10	484
11-3/4 - 13-3/8	11	8-5/8	6-5/8	12	597
13-1/2 - 15	12-3/4	9-15/16	6-5/8	14	794

Dimensions in inches, weights in lb (approximate). Hole size is the range the tool is designed for; max. fish Ø applies to the mill shoe.



DESIGN FEATURES

- **Reverse circulation:** a steel ball dropped down the string seats in the valve and sends flow through the barrel passages, out against the hole wall and back up the barrel.
- **Junk catcher:** spring-loaded fingers fold back to admit core and fish, then dig in on pulling to cut off the core.
- **Mill shoes:** type A with side wings (standard); type B hard-faced with sintered tungsten carbide; type C flat-bottomed, hard-faced, for cutting cores.
- **Finger shoe:** for loose or larger junk; fingers close beneath the fish; replaceable fingers.
- **Magnet insert:** fits the catcher recess, converts to a fishing magnet and keeps reverse circulation.

OPERATING PARAMETERS

Before dropping ball	Circulate a few minutes about 10 ft off bottom
Mill shoe	Cut a core of at least 10 in.; 1 ton on hard-faced shoes, 50 - 150 rpm
Finger shoe	Rotate slowly, lower to about 2 tons on bottom
Precaution	String tool-joint IDs must pass the steel ball; keep the ball at surface while running in

Add a junk sub above the top sub if small junk is prevalent. Wash the tool after each job, paying attention to the space between inner and outer barrels.

HOW TO ORDER

Please specify: 1. tool name and type (standard or W7R) 2. OD and type of shoe 3. top connection 4. recommended spares: 2 junk catchers, 2 mill shoes, 1 valve cup, 1 valve seat, 1 steel ball 5. accessories: 2 finger shoes, 4 finger shoe replacements, 1 magnet insert.

JUNK BASKET

JUNK BASKET WITH DOUBLE CATCHERS

FISHING

GENERAL DESCRIPTION

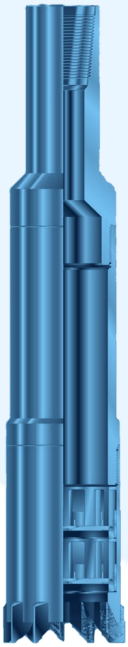
SIMPLE, DEPENDABLE FISHING TOOL FOR RETRIEVING LOOSE JUNK FROM THE BOTTOM OF THE HOLE: BIT CONES, BEARINGS, BROKEN SLIPS, WIRELINE, HAND TOOLS, DEBRIS FROM TWISTED-OFF STRINGS AND MILLING CUTTINGS. TWO FREE-REVOLVING CATCHERS SIT INSIDE A MILL SHOE, SO THE BASKET CUTS A SHORT CORE THAT LOCKS THE JUNK IN THE BARREL. IT CAN ALSO TAKE A CORE SAMPLE, DRILL A FULL-GAUGE HOLE OR REAM. MAGNET INSERTS AND ACCESSORY SHOES ADAPT IT TO THE JOB.

TECHNICAL CHARACTERISTICS

Hole size	Barrel OD	Shoe OD	Max. fish Ø	Teeth	Wt (lb)
3-3/4 - 4-1/8	3-5/8	3-5/8	2-11/16	8	55
4-1/4 - 4-1/2	3-3/4	4-1/16	2-15/16	8	60
4-5/8 - 5	3-7/8	4-1/2	3-9/32	8	65
4-5/8 - 5	3-7/8	4-1/2	3-3/8	8	65
5-1/8 - 5-1/2	4-1/4	4-7/8	3-23/32	8	80
5-3/8 - 6	5-1/8	5-1/8	3-25/32	8	80
5-5/8 - 6	4-3/4	5-3/8	4-1/16	8	100
6-1/8 - 6-1/2	5-3/4	5-3/4	4-13/32	8	200
6-1/8 - 6-1/2	5-1/8	5-7/8	4-5/16	8	110

Hole size	Barrel OD	Shoe OD	Max. fish Ø	Teeth	Wt (lb)
6-1/8 - 6-1/2	5-1/4	5-7/8	4-1/2	8	110
6-5/8 - 7	5-3/4	6-1/4	4-13/16	8	125
7-1/4 - 8	6-1/2	7-1/8	5-7/16	8	160
8-1/4 - 9	7-1/2	8-1/8	6-3/16	8	205
9-1/4 - 10-1/8	8-1/2	9-1/8	7-3/16	10	255
10-1/4 - 11-5/8	9-3/8	10-1/8	8-1/16	12	290
11-3/4 - 12-1/2	10-3/8	11-1/4	9-1/16	12	360
12-5/8 - 15	11-3/8	12-1/4	10-1/16	12	450
15 - 20	13-3/4	14-1/2	12-1/16	16	655

Dimensions in inches, weights in lb (approximate). Shoes should be at least 1/8 in. over the barrel size.



DESIGN FEATURES

- **Double catchers:** the upper fingers reach half way to the centre and break the core on pull; the lower catcher has alternating long and short fingers forming a close-fitting basket.
- **Free-revolving catchers:** spin inside the shoe, which prevents finger breakage.
- **Rivet-free catchers:** manganese bronze or steel fingers, redressed easily on location.
- **Magnet insert:** replaces the two catchers and turns the basket into a fishing magnet, useful in hard formations or to clean the hole before diamond drilling.

SHOE SELECTION

- **Type A mill shoe:** side wings, hard metal facing; standard, for soft formations and loose fish.
- **Type B mill shoe:** hard-faced with sintered tungsten carbide, for hard formations.
- **Type C mill shoe:** flat bottom, hard-faced on bottom, inner and outer faces; cuts cores and burns metal.
- **Finger shoe:** for large, loose junk or junk too big for the catchers.

OPERATING PARAMETERS

Approach	Start circulation at reduced pressure just before bottom, rotate and lower slowly onto the fish
Mill shoes	1 ton on bottom, 50 - 150 rpm
Finger shoes	Rotate and lower slowly to about 2 tons on bottom
Breaking the core	Stop rotation and circulation, release torque, take an upward pull; a hydraulic jar above helps

After use, disassemble, clean and lubricate all internal parts, inspect catcher fingers and springs, and paint or grease the outside.

HOW TO ORDER

Please specify: 1. tool name and size 2. OD and type of shoe 3. top connection, if other than standard 4. recommended spares: 2 upper catchers, 2 lower catchers, 3 mill shoes, 2 finger shoes, 4 finger shoe replacements.

HEC

HYDRAULIC EXTERNAL CUTTER

FISHING

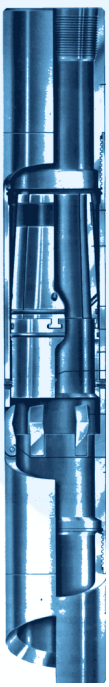
GENERAL DESCRIPTION

HYDRAULICALLY ACTUATED EXTERNAL CUTTER THAT CUTS AND RECOVERS TUBING OR DRILL PIPE IN A SINGLE TRIP. RUN ON A WASHOVER STRING, IT CUTS STUCK PIPE FROM THE OUTSIDE: THE KNIVES ARE FED BY PUMP PRESSURE, GIVING FAST, SMOOTH AND CLEAN CUTS UNDER SENSITIVE OPERATOR CONTROL. IT CUTS ALL TYPES OF DRILL PIPE AND TUBING, REGARDLESS OF TOOL-JOINT STYLE.

TECHNICAL CHARACTERISTICS

Cutter ref.	Pipe to cut	Cutter O.D.	Cutter I.D.	Max. size passed over*	Min. hole to run in	Wt (lb)
70086	1.315 - 2-1/16	3-3/4	2-7/8	2-3/4	4	55
34415	1.315 - 2-3/8	4-1/16	3-3/16	3-1/16	4-5/16	60
70209	1.900 - 2-7/8	4-7/16	3-5/8	3-7/16	4-11/16	73
41727	1.900 - 2-7/8	4-11/16	3-7/8	3-3/4	4-15/16	85
34551	2-1/16 - 4	5-5/8	4-5/16	4-3/16	5-7/8	95
51148	2-3/8 - 4	6-1/16	4-7/8	4-13/16	6-1/4	105
35957	3-1/2 - 5	8	6-1/2	6-3/8	8-1/2	230

* Maximum pipe O.D. the cutter will pass over, dressed with the largest piston segment. Dimensions in inches, weights in lb (approximate).



DESIGN FEATURES

- **Cuts and recovers:** tubing or drill pipe in one trip, inside a washover string.
- **Hydraulic feed:** pump pressure drives the knives; sensitive control of the cut.
- **Run-in protection:** shear pins hold the piston until the cut starts (about 145 psi).
- **Interchangeable piston segments:** one cutter covers several pipe sizes; threaded-edge segments pass collars.
- **Cool, flushed cut:** each segment has a fluid passage for flushing and cooling.
- **Hardened knives:** tempered tool steel, replaceable in the field.
- **Guide options:** standard cut-lip, extra long wall hook or mill-toothed.

OPERATING NOTES

- **Before the run:** wash over the stuck pipe to centralise it at the cut point.
- **Avoid surging:** pressure surges can break a knife tip.
- **After the cut:** torque drops and the weight indicator rises; raise the string slowly to confirm.

PERFORMANCE PARAMETERS

Rotary speed while cutting	15 - 25 rpm
Shear pins	1 pin: about 377 lb; 2 pins: 754 lb
Proving the cut	Torque drop, weight rise
Cut position	1 joint above washover shoe

PULL CAPACITY AND HYDRAULICS

Cutter ref.	Max. load (lb)	psi	gpm
70086	2,530 - 7,910	20 - 40	120 - 220
34415	1,860 - 3,340	20 - 45	120 - 220
70209	1,550 - 3,570	10 - 25	120 - 145
41727	1,128 - 1,780	10 - 30	100 - 200
34551	6,100 - 16,800	15 - 55	180 - 230
51148	2,350 - 4,800	15 - 40	125 - 200
35957	5,160 - 9,040	10 - 20	140 - 180

Approximate values, depending on pipe size. Static loads: if shock loaded, reduce by 50%.

HOW TO ORDER

Please specify: 1. cutter ref. and size of pipe to cut 2. piston segment size 3. connection (top and bottom) 4. guide type (standard, wall hook or mill-toothed) 5. recommended spares: knives, knife pins, shear pins, piston and rubber rings.

PIPE CUTTER

INTERNAL HYDRAULIC CASING CUTTER

CUTTING

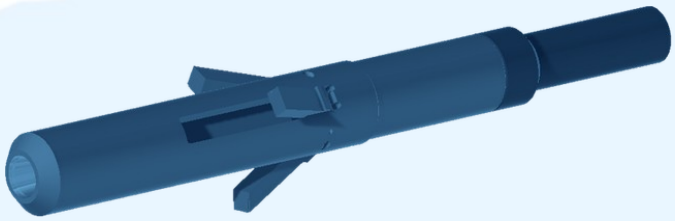
GENERAL DESCRIPTION

INTERNAL HYDRAULIC CUTTER THAT CUTS SINGLE OR MULTIPLE STRINGS OF CONCENTRIC OR ECCENTRIC CASING. IT CAN BE USED IN ANY CUTTING APPLICATION WHERE ROTATION AND PRESSURE CAN BE APPLIED TO THE TOOL. IT IS MAINLY RUN IN WELL ABANDONMENT OPERATIONS, WHEN CASINGS HAVE BEEN CEMENTED AND CANNOT BE CUT AND PULLED INDIVIDUALLY.

TECHNICAL CHARACTERISTICS

Designed to cut (in)	4-1/2 - 5-1/2	6-5/8 - 13-3/8	9-5/8 - 16	13-3/8 - 36
Top connection (API)	2-3/8 IF	3-1/2 REG	6-5/8 REG	6-5/8 REG or 7-5/8 REG
Tool diameter (in)	3-5/8	5-3/4	8-1/4	11-3/4

Dimensions in inches. Cutter arms are available in various sizes to cut single or multiple strings.



DESIGN FEATURES

- **Casing cutting:** single or multiple strings, concentric or eccentric.
- **Hydraulic and rotary:** works wherever rotation and pressure can be applied to the tool.
- **Abandonment duty:** cuts cemented casings that cannot be cut and pulled one by one.
- **Carbide-dressed arms:** crushed carbide on the cutter arms, in several sizes for single or multiple strings.
- **Three knife reloading types:** standard (general use), hardfacing (reinforced, high-strength steels) and carbide (high wear resistance, abrasive environments).

OPERATING PARAMETERS

Flow rate	Up to 400 L/min
Head pressure	35 to 60 bar
Maximum	80 bar / 90 rpm
Surface test	300 L/min at 5 bar
Start of rotation	70 rpm (max. 100)
Start of pumping	150 L/min

If pressure rises too quickly in the surface test, the piston may be blocked.

CUTTING PROCEDURE

- **Record parameters:** PUW, SOW and free torque.
- **Prepare:** close the purge line and reset the WOB.
- **Start:** begin rotation, then pump at 150 L/min while monitoring return flow and torque.
- **Cut:** once return is confirmed, increase gradually to 50 - 60 bar, then stabilise the flow without further pump adjustment.

HOW TO ORDER

Please specify: 1. casing size range to cut 2. top connection 3. knife reloading type (standard, hardfacing or carbide) 4. single or multiple strings to cut.

SECTION MILL

HYDRAULICALLY ACTUATED CASING SECTION MILL

MILLING

GENERAL DESCRIPTION

HYDRAULICALLY ACTUATED TOOL THAT MILLS A SECTION OUT OF CASING OR TUBING, AND CAN ALSO BE USED TO CUT A SINGLE STRING OF CASING. PUMP PRESSURE ACROSS THE PISTON PUSHES A CAM DOWN AND EXPANDS THE KNIVES AGAINST THE CASING: THE CUT-OUT KNIVES PART THE PIPE FIRST, THEN ALL THE KNIVES TAKE PART IN MILLING. WHEN CIRCULATION STOPS, THE PISTON SPRING RETRACTS THE CAM AND THE KNIVES FOLD BACK INTO THE BODY FOR RETRIEVAL. KNIVES ARE DRESSED WITH TUNGSTEN CARBIDE INSERTS, SUITED TO POORLY CEMENTED, SPLIT OR CORRODED CASING; A SECTION OF UP TO 60 FT IS TYPICALLY MILLED WITH ONE SET OF KNIVES.

TECHNICAL CHARACTERISTICS

Tool series	Casing size	Body Ø	Fishing neck length	Fishing neck Ø	Overall length	Top pin API Reg.	Wt (lb)
3600	4-1/2	3-5/8	18	3-1/8	56	2-3/8	135
4100	5	4-1/8	18	3-1/4	66	2-3/8	175
4500	5-1/2, 6	4-1/2	18	4-1/8	70	2-7/8	220
5500	6-5/8, 7	5-1/2	18	4-3/4	74	3-1/2	350
6100	7-5/8	6-1/8	18	4-3/4	74	3-1/2	368
7200	8-5/8, 9-5/8	7-1/4	18	5-3/4	89	4-1/2	554
8200	9-5/8	8-1/4	18	5-3/4 or 8	87	4-1/2 or 6-5/8	900
9200	10-3/4, 11-3/4	9-1/4	18	5-3/4 or 8	87	4-1/2 or 6-5/8	980
11700	13-3/8, 16	11-1/2	18	8 or 9	90	6-5/8 or 7-5/8	1,725

Dimensions in inches, weights in lb (approximate). Standard API regular pin connections; others on request.

DESIGN FEATURES

- **Staged knife opening:** three knives cut out, then all six mill.
- **Carbide cutting structure:** inserts laid in a brick-work pattern for footage and penetration.
- **Positive cut-out signal:** the Flo-Tel option doubles the flow area once the casing is parted, giving a 200 - 250 psi pressure drop at surface. Standard on series 4100 to 11700; not available on 3600.
- **Self-retracting:** piston spring closes the knives when pumps stop.
- **Jetted top sub:** limits velocity through the piston above 300 gpm.
- **Simple servicing:** replaceable knives, arm-stop stabilizers and Flo-Tel unit.

FLOW RATES (GPM)

Tool series	During cut-out	During milling
3600	110 - 160	110 - 160
4100	80 - 125	110 - 160
4500	80 - 125	150 - 200
5500	80 - 125	200 - 250
6100	80 - 125	200 - 250
7200	80 - 125	200 - 300
8200	80 - 125	300 - 400
9200	80 - 125	350 - 450
11700	350 - 450	350 - 600

OPERATING PARAMETERS

Rotary speed, cut-out	60 rpm to start
Rotary speed, milling	80 - 120 rpm
Weight on mill	4,000 - 8,000 lb
Min. pressure at tool	about 300 psi to cut out
Pipe cutter use	80 - 100 rpm

HOW TO ORDER

Please specify: 1. tool series 2. size and weight of casing to be milled 3. top pin connection.

RTU

ROCK-TYPE UNDERREAMER

ENLARGEMENT

GENERAL DESCRIPTION

RUGGED THREE-CONE UNDERREAMER THAT ENLARGES A HOLE TO NEARLY TWICE ITS OWN BODY DIAMETER. A WIDE RANGE OF CONE TYPES MATCHES THE CUTTER TO THE FORMATION, AND A CHOICE OF ORIFICE SIZES TAILORS PERFORMANCE TO THE RIG HYDRAULICS. THE TOOL CAN BE SERVICED ON LOCATION, THE CUTTER ARMS ARE CHANGED QUICKLY ON THE RIG FLOOR, AND FULL CIRCULATION IS KEPT AT ALL TIMES WITH WATER, MUD, AIR, AERATED MUD OR FOAM.

TECHNICAL CHARACTERISTICS

Tool series	Std. opening Ø	Optional opening Ø	Body Ø	Collapsed Ø	Through casing Ø x lb/ft	Fishing neck length	Fishing neck Ø	Overall length	Top pin API Reg.	Wt (lb)
3600	6	4-3/4 - 6-1/2	3-5/8	3-3/4	4-1/2 x 15	8	3-3/8	26-1/2	2-3/8	175
4500	6-1/2, 8-1/2	6 - 9	4-1/2	4-5/8	5-1/2 x 20	18	4-1/8	67	2-7/8	235
5700	11	8 - 11	5-3/4	5-7/8	7 x 38	18	4-3/4	76-1/2	3-1/2	380
5800	11	8 - 11	5-7/8	5-7/8	6-5/8 x 20	18	4-3/4	76-1/2	3-1/2	380
6000	12	11 - 12	6	6-1/8	7 x 26	18	4-3/4	78-1/2	3-1/2	380
6100	12	11 - 12	6-1/8	6-1/8	7 x 20	18	4-3/4	78-1/2	3-1/2	380
6200	12	11 - 13	6-1/4	6-1/4	7 x 17	18	4-3/4	78-1/2	3-1/2	380
7200*	14	9 - 14	7-1/4	7-3/8	8-5/8 x 40	18	5-3/4	86	4-1/2	775
8200*	16	10 - 16	8-1/4	8-3/8	9-5/8 x 47	18	5-3/4 or 8	89	4-1/2 or 6-5/8	920
9500*	17-1/2	13 - 18	9-1/2	9-3/4	10-3/4 x 45	18	8	91	6-5/8	1,160
11700*	17-1/2	14-3/4 - 22	11-3/4	12-1/4	13-3/8 x 68	20	8	91	6-5/8	1,670
15000 LP*	26	17-1/2 - 30	14-3/4	14-3/4	16 x 75	20	8 or 9	97	6-5/8 or 7-5/8	2,800
22000	32 - 40	32 - 40	22	22	24-1/2 x 113	20	9 or 10	124-1/4	7-5/8 or 8-5/8	5,900

* Available with PDC Bearclaw cutters. Dimensions in inches, weights in lb (approximate). Optional opening diameters on request.



DESIGN FEATURES

- **Three-cone cutters:** enlarges to almost twice the body diameter.
- **Cutter options:** milled tooth, TCI or PDC Bearclaw cutting structures.
- **Tailored hydraulics:** orifice sizes selected to suit the rig conditions.
- **Full circulation:** constant flow volume through the tool.
- **Field serviceable:** serviced on location; arms swapped on the rig floor.
- **Optional Flo-Tel:** pump pressure drops 200 - 250 psi once the arms are fully extended, confirming full gauge and avoiding second trips.

OPERATING NOTES

- **Running in:** go slowly at the top of the section; surge pressure can open the arms and damage casing.
- **Rate of penetration:** keep below 100 ft/hr or the hole may not reach drift gauge.
- **Dull cutters:** pull at the first sign of dulling.

PERFORMANCE PARAMETERS

Rpm to shoulder depth	30 - 50 rpm
Rpm, cutting shoulder	80 - 150 rpm max.
Rpm, underreaming	80 rpm
Weight on tool	1,000 lb per in. of body Ø
With Flo-Tel	80 - 150 rpm, max. pump pressure
Circulating medium	Water, mud, air, foam

CUTTING THE SHOULDER

- **Start:** establish flow, rotate at 80 - 150 rpm and drill off slowly for 5 - 10 minutes.
- **Open the arms:** lift off with the pump on and repeat to be sure they are fully open.
- **Shoulder reached:** weight starts to register, then add weight to underream.

HOW TO ORDER

Please specify: 1. tool series 2. top and bottom connections 3. fishing neck Ø 4. opening Ø 5. size and weight of casing to run through, if known 6. bullnose or bits (ordered separately) 7. cutting structure (milled tooth, TCI or PDC).

DTU

DRILLING-TYPE UNDERREAMER

ENLARGEMENT

GENERAL DESCRIPTION

DRILLING-TYPE UNDERREAMER THAT ENLARGES PREVIOUSLY DRILLED PILOT HOLES. A BOTTOM BOX CONNECTION LETS A BIT OR A BULLNOSE RUN BELOW THE TOOL, SO THE DTU CAN DRILL AND UNDERREAM AT THE SAME TIME. MUD FLOW CAN BE SPLIT BETWEEN THE CUTTER ARMS AND THE BIT OR BULLNOSE. IT RUNS WITH WATER, MUD, AIR, AERATED MUD OR ANY OTHER CIRCULATING MEDIUM.

TECHNICAL CHARACTERISTICS

Tool series	Std. opening Ø	Optional opening Ø	Body Ø	Collapsed Ø	Through casing Ø x lb/ft	Fishing neck length	Fishing neck Ø	Overall length	Top pin API Reg.	Wt (lb)
3600	6	5-1/2 - 6	3-5/8	3-5/8	4-1/2 x 15	8	3-3/8	35	2-3/8	170
5700	8-3/4	7 - 8-3/4	5-3/4	6	7 x 38	18	4-3/4	70	3-1/2	360
7200*	11	9 - 11	7-1/4	7-1/2	8-5/8 x 40	18	5-3/4	74	4-1/2	770
8200*	14	10 - 14	8-1/4	8-1/4	9-5/8 x 53	18	5-3/4 or 8	79	4-1/2 or 6-5/8	900
9500*	15	12 - 15	9-1/2	10-1/4	11-3/4 x 71	18	8	82	6-5/8	1,150
11700*	17-1/2	14-3/4 - 20	11-3/4	11-3/4	13-3/8 x 92	20	8	96	6-5/8	1,670
15000*	17-1/2 - 26	17-1/2 - 26	14-3/4	14-3/4	16 x 75	20	8 or 9	97	6-5/8 or 7-5/8	2,800
17000*	32	24 - 32	17	17	18-5/8 x 78	20	9 or 10	87	7-5/8 or 8-5/8	3,000
22000	36	28 - 36	22	22	24-1/2 x 113	20	9 or 10	100	7-5/8 or 8-5/8	4,400

* Available with PDC Bearclaw cutters. Dimensions in inches, weights in lb (approximate). Optional opening diameters on request.

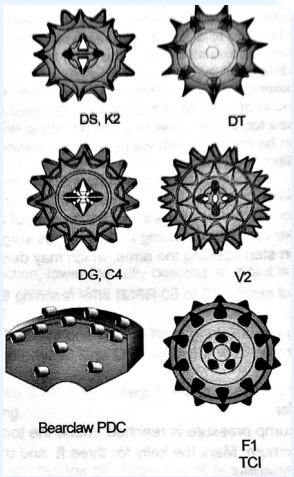
DESIGN FEATURES

- **Drill and underream in one run:** bottom box connection for a bit or bullnose.
- **Flexible hydraulics:** flow can be divided between the arms and the bit or bullnose; orifice jets are selected to suit the available hydraulics.
- **Cooling and cleaning:** jet placement keeps the cutters cool and aids annular lift of cuttings.
- **Extra flow options:** jetted bullnose and jetted top sub to divert more flow when needed.
- **Cutting structures:** open-bearing milled tooth (standard), TCI or PDC Bearclaw.

PERFORMANCE PARAMETERS

Rotary speed, reaching shoulder depth	30 - 50 rpm
Rotary speed, cutting the shoulder	80 - 150 rpm max.
Rotary speed, underreaming	80 rpm
Weight on tool	1,000 lb per inch of body Ø
Stabilization	Full-gauge stabilizer 60 - 90 ft above and below
Position in the BHA	Middle of the drill collars

CUTTER OPTIONS



Gauge diameter tolerances, arms open

Size (in)	Tolerance
6 - 9	+1/16 - 0
9-1/2 - 13	+3/32 - 0
14 - 18	+1/8 - 0
20 - 30	+3/16 - 0

Applies to arm sets of RTU, DTU, drag and XTU underreamers, all cutting structures.

HOW TO ORDER

Please specify: 1. tool series 2. top and bottom connections 3. fishing neck Ø 4. opening Ø 5. size and weight of casing to run through, if known 6. bullnose or bits (ordered separately) 7. cutting structure (milled tooth, TCI or PDC).

HOLE OPENER

FIXED-DIAMETER HOLE OPENER

ENLARGEMENT

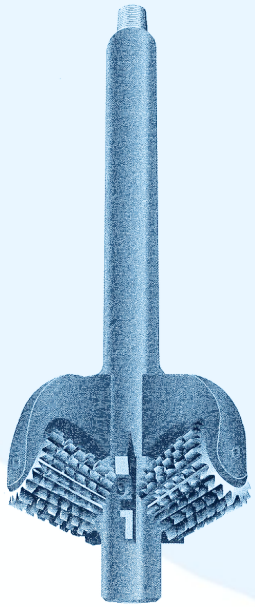
GENERAL DESCRIPTION

FIXED-DIAMETER HOLE OPENER THAT ENLARGES A PREVIOUSLY DRILLED PILOT HOLE FROM THE SURFACE. THE CUTTERS ARE RIGIDLY ATTACHED TO THE BODY: THERE IS NO HYDRAULIC ACTUATION, AND THE TOOL ALWAYS CUTS AT ITS FULL GAUGE. USED MAINLY FOR SURFACE HOLES AND CONDUCTOR CASING, WHERE A BIGGER HOLE GIVES ENOUGH CLEARANCE TO RUN AND CEMENT THE CASING (FOR EXAMPLE, A 17-1/2 IN. HOLE OPENED TO RUN 13-3/8 IN. CASING). THE CHOICE OF CUTTERS LETS IT HANDLE SOFT TO HARD-ABRASIVE FORMATIONS. CUTTER SIZES FROM 26 TO 42 IN.

TECHNICAL CHARACTERISTICS

Std. opening Ø	Min. pilot hole Ø	Fishing neck length	Fishing neck Ø	Top pin API Reg.	Bottom neck length	Bottom neck Ø	Bottom box API Reg.	Body length	Min. bore Ø
26	14	60	9-1/2	6-5/8 - 7-5/8	12	9-1/2	7-5/8	96	1
28	16	60	9-7/8	6-5/8 - 7-5/8	12	9-1/2	7-5/8	96	1
30	18	60	9-7/8	6-5/8 - 7-5/8	12	9-1/2	7-5/8	100	2
32	20	60	10	7-5/8 - 8-5/8	15	9-1/2	8-5/8	100	2
34	22	60	10	7-5/8 - 8-5/8	15	9-1/2	8-5/8	101	2
36	24	60	10	7-5/8 - 8-5/8	15	9-1/2	8-5/8	103	3
38	26	60	10	8-5/8	15	9-1/2	8-5/8	106	3
40	28	60	10	8-5/8	15	9-1/2	8-5/8	106	3
42	30	60	10	8-5/8	15	9-1/2	7-5/8	106	3

Dimensions in inches, approximate. Intermediate sizes or sizes larger than 42 in. available on request.



DESIGN FEATURES

- **Fixed gauge:** cutters rigidly mounted; no hydraulic actuation.
- **Demountable cutters:** easily replaced on the rig floor.
- **Cutters:** sealed-bearing milled tooth or tungsten carbide insert.
- **Long fishing necks:** can be made up below the rotary table despite the large size.
- **Replaceable nozzles:** effective hole cleaning.
- **Wide formation range:** soft to hard-abrasive.

NOZZLES AND CUTTERS

- **Nozzles:** threaded retainer holds the jet nozzle in its sleeve; O-ring seal.
- **Cutter sets:** replacement sets include all parts needed.
- **Rebuild kits:** cutter bearing rebuild kits on special order.

OPERATING PARAMETERS

Formation	Cutter	Weight (lb)	rpm
Soft	Milled tooth	15,000 - 25,000	125 - 150
Medium	Milled tooth	20,000 - 35,000	75 - 85
Hard	Button type	35,000 - 45,000	50 - 60
Extr. hard	TCI button	30,000 - 45,000	30 - 40

Guideline for 19-3/4 - 26 in. holes; smaller sizes need less weight.

GAUGE TOLERANCE (ARMS OPEN)

Size (in)	Plus	Minus
18 - 26	+1/8	- 1/16
27 - 42	+5/32	- 1/16
43 and larger	+3/16	- 1/16

Gauge as dressed, not ground; milled tooth and TCI.

MASTER DRILLER (CONE ARM VERSION)

Cone-arm hole opener, up to 36 in.; one body takes several arm sizes. Bottom box for bit or bullnose. Milled tooth, TCI or PDC cones; soft to medium-hard formations. Body series 8200 to 15000: std. opening 12-1/4 to 36-5/8 in., min. pilot hole 8-1/2 to 24-5/8 in., body Ø 8-1/4 to 15-5/8 in., 640 - 1,900 lb.

HOW TO ORDER

Please specify: 1. pilot hole size 2. hole opening size 3. top and bottom connections if not standard 4. fishing neck diameter if not standard 5. cutter type (milled tooth or tungsten carbide insert) and formation (soft or medium).

SPX

DRAG-TYPE UNDERREAMER, PDC CUTTERS

ENLARGEMENT

GENERAL DESCRIPTION

DRAG-TYPE UNDERREAMER WITH SPX PDC CUTTERS ON THE CUTTING EDGE: THE HARDNESS AND WEAR RESISTANCE OF POLYCRYSTALLINE DIAMOND, BACKED BY THE STRENGTH OF CEMENTED TUNGSTEN CARBIDE. THE CUTTERS SHEAR THROUGH SOFT TO MEDIUM-HARD FORMATIONS FASTER THAN CONVENTIONAL TOOLS AND NEED LESS TOOL WEIGHT. USED IN POORLY CONSOLIDATED SOFT TO MEDIUM FORMATIONS WHERE LARGER DIAMETERS ARE REQUIRED FOR GRAVEL PACKING OR CEMENTING. A PILOT HOLE CAN BE ENLARGED UP TO THREE TIMES THE BODY DIAMETER IN A SINGLE TRIP.

TECHNICAL CHARACTERISTICS

Tool series	Body Ø	Min. pilot hole Ø	Std. expanded Ø, SPX (PDC)	Std. expanded Ø, tungsten carbide	Fishing neck length	Fishing neck Ø	Overall length	Top pin API	Wt (lb)
3600	3-3/8	3-3/4	N/A	9	8	3-3/8	26-1/2	2-3/8 Reg.	tbc
4500	4-1/2	4-3/4	6-1/2, 6-3/4, 8-1/2	Upon request	18	4-1/4	69	3-1/2 IF	230
4700	4-3/4	5	Upon request	12	18	4-1/8	67	2-7/8 Reg.	tbc
5700	5-3/4	6	7-1/2, 8, 8-1/2, 12, 13	16	18	4-3/4	70	3-1/2 Reg.	350
7200	7-1/4	7-5/8	10, 12-1/4, 13, 14, 15, 16	22	18	5-3/4	78	4-1/2 Reg.	750
8200	8-1/4	8-1/2	10, 12-1/4, 14, 15, 16, 17	23	18	5-3/4 or 8	78	4-1/2 or 6-5/8 Reg.	tbc
9500	9-1/2	9-7/8	12-1/4, 17-1/2	28	18	8	78	6-5/8 Reg.	tbc
11700	11-3/4	12-1/4	Upon request	36	18	8 or 9	86	6-5/8 or 7-5/8 Reg.	tbc

All dimensions in inches, weights in lb (approximate). Other expanded diameters, orifices and connections available on request. tbc: to be confirmed.

DESIGN FEATURES

- **Self-cleaning arms:** a circulation jet nozzle sends part of the flow out of each arm pocket, cleaning the cutting surfaces, improving cuttings removal and dissipating frictional heat.
- **Durable arms:** dressed with long-wearing cutting-grade tungsten carbide.
- **Field serviceable:** low-cost cutter arms and orifices can be replaced in the field.
- **Optional Flo-Tel:** pump pressure drops by 200 - 250 psi when the arms are fully extended, confirming full-gauge underreaming.

PERFORMANCE PARAMETERS

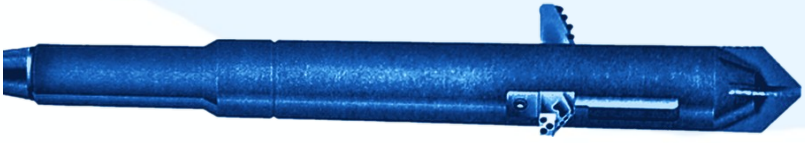
Rotary speed, cutting the shoulder	40 - 60 rpm
Flow rate	Increase gradually, 250 - 450 gpm
Weight on tool	Start at 3,000 lb, up to 10,000 lb
With Flo-Tel	80 - 150 rpm at max. pump pressure
Circulating medium	Water, mud, air, aerated mud, foam
Max. enlargement	3 x body Ø in a single trip

ADVANTAGES

- Faster penetration and more time on bottom
- Less rig time and lower total cost per foot
- Quick drilling through producing zones limits formation damage and hole-stability problems

HOW TO ORDER

Please specify: 1. tool series 2. top and bottom connections 3. fishing neck Ø 4. opening Ø 5. size and weight of casing to run through, if known 6. bullnose or bits (ordered separately) 7. cutting structure (milled tooth, TCI or PDC).



INTENSIFIER

JAR INTENSIFIER, HYDRAULIC FLUID SPRING

JARRING

GENERAL DESCRIPTION

HYDRAULIC FLUID SPRING RUN WITH AN OIL JAR TO BOOST JARRING. AS THE STRING IS PULLED, A PISTON COMPRESSES A HIGHLY COMPRESSIBLE SILICONE FLUID AND STORES ENERGY JUST ABOVE THE DRILL COLLARS. WHEN THE JAR TRIPS, THIS ENERGY ACCELERATES THE COLLARS AND JAR UPWARD FOR A HIGH-IMPACT BLOW, LARGELY INDEPENDENT OF STRING DRAG OR DEPTH. IT ALSO CUSHIONS THE REBOUND AFTER EACH STROKE, AND ALLOWS FEWER DRILL COLLARS, ESPECIALLY IN SHALLOW HOLES.

TECHNICAL CHARACTERISTICS

O.D.	I.D.	Total stroke to solid	Collar weight (lb)	Tensile at yield (lb)	Torque at yield (ft-lb)	Fluid volume (gal)
1-5/8	1/4	6	1,100 - 1,400	43,200	260	0.13
1-13/16	5/16	6	1,360 - 1,800	59,400	340	0.195
2-1/4	3/8	6	1,560 - 2,100	118,500	1,800	0.211
2-29/32	1	12-3/4	2,200 - 3,000	194,800	2,260	0.692
3-1/8	1	8-3/4	2,400 - 3,300	229,200	4,068	0.375
3-3/4	1-1/4	8-1/4	4,200 - 5,700	345,000	7,640	0.82
3-3/4	1-1/2	7-7/8	3,400 - 4,600	299,700	5,340	0.63
3-3/4	1-7/8	7-5/8	3,500 - 4,700	179,500	2,980	0.613
4-1/4	1-15/16	8-5/8	3,500 - 4,700	430,300	9,920	0.92
4-1/2	2-3/8	10-3/8	3,600 - 4,900	375,000	11,160	1.15
4-3/4	1-1/2	8-7/8	6,300 - 8,500	591,900	18,420	1.0
4-3/4	2	10-1/8	5,600 - 7,500	468,800	17,200	1.35
6	2	8-5/8	10,200 - 13,800	937,000	34,320	1.57
6-1/4	2-1/4	13	11,800 - 16,000	917,400	40,680	4.24
6-3/4	2-3/8	13	13,000 - 17,500	1,013,800	48,660	3.45
7-3/4	3-1/16	13	11,000 - 15,000	1,587,900	64,020	4.65
9	3-3/4	13	12,000 - 16,000	1,621,000	92,260	3.2

Dimensions in inches. Each intensifier is matched to a jar of the same O.D. x I.D.; connections as for the matching jar. Tensile strengths are calculated theoretical yield values (about +/-20%); torque at yield is theoretical. Figures are a guide only: apply a safety factor.

DESIGN FEATURES

- **Energy storage:** compressible silicone fluid acts as a spring above the collars.
- **Splined mandrel:** torque transmitted in either direction, open or closed.
- **High-pressure seals:** non-extrusion ring assemblies limit seal damage and wear.
- **Safe to fill and use:** fluid-filled, charged with a low-pressure hand pump only.
- **Shock cushioning:** absorbs rebound of the string after each jarring stroke.

OPERATING NOTES

- **String order from the fish:** fishing tool, jar, drill collars, intensifier, string.
- **Stroke:** 6 to 13 in under pull, depending on size.
- **No heavy weights above:** no collars or heavy-weight for at least 1,000 ft above the intensifier.
- **Hot holes:** above 180 F (82 C) bottom-hole temperature, bleed a little fluid before running.

HOW TO ORDER

Please specify: 1. intensifier size and the jar it matches 2. connections and O.D., if not standard 3. spares: packing sets, piston assemblies, seal rings, fill plugs, service kit 4. special intensifier fluid.

JAR Z

HYDRAULIC OIL JAR, STRAIGHT PULL

JARRING

GENERAL DESCRIPTION

STRAIGHT-PULL HYDRAULIC OIL JAR FOR FISHING, DRILL STEM TESTING AND CORING. A METERING PISTON DELAYS THE STROKE WHILE THE OPERATOR TAKES STRETCH IN THE STRING, SO THE BLOW CAN BE SET FROM VERY LIGHT TO VERY HIGH IMPACT WITH NO ADJUSTMENT BEFORE OR AFTER THE RUN. THE HYDRAULIC SYSTEM IS CLOSED: OIL CANNOT ESCAPE AND WELL FLUIDS CANNOT ENTER. THE JAR RESETS WITH ONLY A FEW HUNDRED POUNDS OF WEIGHT AND TRANSMITS FULL TORQUE IN BOTH DIRECTIONS AT ALL TIMES.

TECHNICAL CHARACTERISTICS

Connection	O.D.	I.D.	Total stroke	Test pull load (lb)	Max. jarring load (lb)	Tensile at yield (lb)	Torque at yield (ft-lb)	Collar weight above jar (lb)
Sucker rod	1-5/8	1/4	7-1/16	5,000 - 8,500	15,400	46,300	260	1,100 - 1,450
1-13/16 WFJ	1-13/16	5/16	7	6,000 - 10,000	18,000	59,400	340	1,360 - 1,800
1-1/4 Reg	2-1/4	3/8	6-15/16	7,200 - 12,000	21,000	118,500	1,800	1,560 - 2,100
2-3/8 PH-6	2-29/32	1	11-7/16	12,600 - 19,000	35,400	194,800	2,260	2,200 - 3,000
2-3/8 EUE	3-1/16	1-1/2	11-9/16	12,000 - 16,000	27,800	160,200	2,200	2,300 - 3,100
2-3/8 Reg	3-1/8	1	11-13/16	15,000 - 23,000	32,400	229,200	4,060	2,400 - 3,300
2-7/8 Reg	3-3/4	1-1/4	11-13/16	18,000 - 26,000	56,500	345,000	7,640	4,200 - 5,700
2-3/8 IF	3-3/4	1-1/2	12-1/8	15,000 - 26,000	46,000	299,700	5,340	3,400 - 4,600
2-3/8 EUE	3-3/4	1-7/8	10-1/8	15,000 - 26,000	46,500	179,500	2,980	3,500 - 4,700
2-7/8 IF	4-1/4	1-15/16	11-13/16	18,000 - 28,000	46,700	430,300	9,920	3,500 - 4,700
2-7/8 EUE	4-1/2	2-3/8	12-1/4	18,000 - 28,000	49,000	375,000	11,160	3,600 - 4,900
3-1/2 FH	4-3/4	1-1/2	11-13/16	28,000 - 45,000	85,000	591,900	18,420	6,300 - 8,500
3-1/2 FH	4-3/4	2	13-5/16	28,000 - 45,000	74,500	468,800	17,200	5,600 - 7,500
4-1/2 FH	6	2	11-7/16	41,000 - 62,000	136,400	937,000	34,320	10,200 - 13,800
4-1/2 IF	6-1/4	2-1/4	14-3/8	55,000 - 80,000	159,000	917,400	40,680	11,800 - 16,000
5-1/2 Reg	6-3/4	2-3/8	14-11/16	62,000 - 110,000	172,800	1,013,800	48,660	13,000 - 17,500
6-5/8 Reg	7-3/4	3-1/16	14-3/8	63,000 - 100,000	149,000	1,587,900	64,020	11,000 - 15,000
7-5/8 Reg	9	3-3/4	13-13/16	80,000 - 120,000	214,000	1,621,000	92,260	14,300 - 19,600

Dimensions in inches. Tensile strengths are calculated theoretical yield values (about +/-20%); torque at yield is the theoretical value, recommended operating torque is 50% of it. Figures are a guide only: apply a safety factor.

DESIGN FEATURES

- **Adjustable blow:** impact set by the stretch taken in the string, no prior adjustment.
- **Closed hydraulic system:** internal parts are constantly lubricated; no leakage in or out.
- **Easy reset:** large piston ports open on closing, so only friction weight is needed.
- **Full torque:** transmitted in either direction at any point of the stroke.
- **Fast repeat blows:** limited only by the time to raise and lower the string.

OPERATING NOTES

- **Fishing:** run directly below the drill collars; add a jar intensifier about four collars above the jar.
- **Drill stem testing, coring:** 3 to 15 collars above the jar; does not interfere with test tools.
- **Never exceed the maximum jarring load.** Do not tong the threaded joints; tong at least 4 in away.
- **First blow:** can take seconds to minutes, depending on depth, stretch, temperature and hole condition.

HOW TO ORDER

Please specify: 1. jar size (O.D. x I.D.) 2. top and bottom connections, if not standard 3. spares: packing set, piston assembly, non-extrusion and seal protector rings, fill plugs 4. service kit and jar oil.

BUMPER SUB

FISHING BUMPER SUB

JARRING

GENERAL DESCRIPTION

DURABLE FISHING BUMPER SUB MADE OF HEAT-TREATED ALLOY STEEL FOR SUSTAINED BUMPING WITH MINIMAL MAINTENANCE. RUN IN THE STRING JUST ABOVE THE FISHING TOOL OR SAFETY JOINT, IT LETS THE OPERATOR DELIVER SOLID DOWNWARD OR UPWARD BLOWS AT WILL, FOR EXAMPLE TO RELEASE A FISHING TOOL THAT CANNOT PULL THE FISH. A HEXAGONAL MANDREL SLIDES IN A MATCHING BODY, SO TORQUE IS TRANSMITTED OVER THE WHOLE STROKE. IT CAN ALSO ACT AS A FEED-OFF TOOL FOR INTERNAL PIPE CUTTING.

TECHNICAL CHARACTERISTICS

O.D.	I.D.	Tensile at yield (lb)	Torque at yield (ft-lb)
1-13/16	3/8	68,720	1,020
1-29/32	3/8	78,620	850
2-1/4	1/2	172,900	2,440
2-1/2	1/2	160,600	1,940
3-1/8	1-1/2	158,600	3,300
3-1/8	1	224,900	4,200
3-3/8	1	256,000	5,740
3-3/4	1-1/2	316,200	6,000
3-3/4	1-1/4	268,500	9,400
3-3/4	1-7/8	231,000	6,810
4-1/8	1-1/2	370,200	9,270
4-1/4	1-15/16	305,000	9,200
4-1/4	1-1/2	439,000	9,160
4-1/2	2-3/8	336,600	10,240

O.D.	I.D.	Tensile at yield (lb)	Torque at yield (ft-lb)
4-5/8	1-7/8	442,000	13,440
4-5/8	2	465,900	12,280
4-3/4	1-1/2	581,100	13,800
4-3/4	2	465,900	8,200
5-1/2	2	613,400	22,070
5-3/4	2	723,600	26,850
6	2	767,000	29,490
6-1/8	2-1/4	768,600	30,840
6-1/4	2-1/4	864,400	38,000
6-1/2	2-1/4	864,400	42,000
6-3/4	2-3/4	871,000	37,600
7-3/4	3-1/2	1,060,000	50,400
8	3-1/2	1,060,000	54,600
9	4	1,600,000	64,110

Dimensions in inches. Standard stroke 20 in; strokes of 30, 36, 42, 48 and 60 in also available (convert by changing mandrel and middle body only). Tensile strengths are calculated theoretical yield values (about +/-20%); same values typically apply to other strokes. Figures are a guide only: apply a safety factor.

DESIGN FEATURES

- **Solid blows both ways:** downward on closing, upward on opening against the knocker.
- **Hexagonal mandrel:** torque transmitted over the full stroke.
- **Alloy steel body:** heat-treated parts for bumping, tension and torque.
- **Sealed knocker:** packing rings seal mandrel to middle body; optional chevron packing.
- **Low maintenance:** wash and lubricate after use; check seals for wear.

OPERATING NOTES

- **Bump down:** pull to fully open plus stretch, then drop the string and brake before the sub closes for repeated blows, or drop freely for one solid blow.
- **Jar up:** take moderate stretch, drop by that stretch only and brake; the sub rebounds open and strikes the knocker.
- **At surface:** hold a sledge between body and mandrel shoulder, slack off and pull it out to release a fishing tool.

HOW TO ORDER

Please specify: 1. size (O.D. x I.D.) 2. connection size and type 3. length of stroke 4. recommended spares: 8 non-extrusion rings, 8 seal protector rings, 16 knocker seals.

Drilling Jar

DOUBLE-ACTING HYDRAULIC / MECHANICAL DRILLING JAR

JARRING

GENERAL DESCRIPTION

HYDRAULIC-DELAY JAR FOR BOTH UPSTROKE AND DOWNSTROKE, WITH A MECHANICAL LATCH, BUILT FOR COMPLEX DRILLING PROGRAMS. THE DOWNSTROKE DELAY MAKES IT EFFECTIVE IN DIRECTIONAL, HORIZONTAL AND EXTENDED-REACH WELLS. BECAUSE NO BLEED-THROUGH OPERATION IS NEEDED WHEN TAGGING OR PICKING UP OFF BOTTOM, IT CAN BE PLACED IN COMPRESSION IN THE DRILLSTRING. LATCH RELEASE SETTINGS ARE ADJUSTABLE.

TECHNICAL CHARACTERISTICS

Tool O.D. (in)	4-1/4	4-3/4	6-1/4	6-1/2	6-3/4	7	8	9-1/2
Tool O.D. (mm)	108	121	159	165	171	178	203	241
Inside Ø (in)	2	2-1/4	2-1/4	2-1/4	2-1/2	2-1/2	2-13/16	3
Max. hole Ø (in)*	6-1/4	7-7/8	12-1/4	12-1/4	12-1/4	12-1/4	17-1/2	26
Max. jar load (lbf)	86,000	84,000	160,000	160,000	190,000	190,000	300,000	350,000
Tensile yield (lbf)	350,000	362,000	755,000	755,000	823,000	1,000,000	1,000,000	1,225,000
Torsional yield (lbf-ft)**	11,500	11,500	37,500	37,500	44,500	70,000	73,500	95,000
Total stroke (in)	17.5	19	22	22	20	20	21	21
Std up latch (lbf)	35,000	40,000	90,000	90,000	90,000	90,000	95,000	100,000
Max. up latch (lbf)	50,000	55,000	140,000	140,000	140,000	140,000	150,000	155,000
Std down latch (lbf)***	15,750	18,000	40,500	40,500	40,500	40,500	42,750	45,000
Max. down latch (lbf)	22,500	27,000	63,000	63,000	63,000	63,000	67,500	69,800
Pump open area (in²)	5.4	6.5	9.6	9.6	11.0	11.0	14.2	15.9
Length, latched (ft)	22	19.8	22.5	22.5	23.0	23.0	22.5	23.0
Weight (lb)	730	850	2,000	2,100	2,300	2,500	3,100	5,200
Ref.	4525	431	442	428	440	480	411	437

* Hole openers not recommended. ** Rating based on the yield of the body connections, independent of tool-joint connections. *** Down latch setting is 45% of the up latch setting, tolerance +/- 5.



DESIGN FEATURES

- **Delay in both directions:** hydraulic delay with mechanical latch, upstroke and downstroke.
- **Adjustable latch:** release settings can be changed up to the maximum values above.
- **Jet-type metering device** controls the hydraulic delay.
- **Safe on the rig floor:** no unexpected jarring while making connections or drilling.
- **Simple to operate:** no torque needed in the jarring procedure.
- **Built for high friction:** weight can be stacked before firing, even in horizontal sections.
- **Consistent firing:** continuous, repeatable jarring performance.

OPERATING PARAMETERS

Placement in string	Tension, neutral or compression
Delay	Hydraulic, up and down
Torque for jarring	None required
Tagging / picking up	No bleed-through needed
Down latch	45% of up latch, +/- 5
Hole openers	Not recommended

HOW TO ORDER

Please specify: 1. tool O.D. and ref. 2. top and bottom connections 3. up and down latch release settings (standard or adjusted, within the maximum) 4. hole diameter and drilling program (directional, horizontal or extended reach).

ECONOMILL

LOW-COST LIGHT-DUTY MILL

MILLING

GENERAL DESCRIPTION

LOW-COST MILLING TOOL FOR LIGHT-DUTY JOBS, DRESSED WITH TUNGSTEN CARBIDE. IT IS EFFECTIVE FOR MILLING PACKERS, RETAINERS, BRIDGE PLUGS AND CEMENT. IT HAS STANDARD API CONNECTIONS, SO IT IS MADE UP AND BROKEN OUT WITH A STANDARD BIT BREAKER AND NEEDS NO ADDITIONAL SUB. FLUID IS DIRECTED ALONG EACH CUTTING BLADE AND THROUGH THE CENTRE OF THE HEAD FOR CUTTINGS REMOVAL AND COOLING.

TECHNICAL CHARACTERISTICS

Tool series	Dressed diameter (in.)	Top pin conn. (API Reg.)	Recommended make-up torque (ft-lb)	Overall length (in.)	Wt (lb)
3000	3-1/4 - 4-3/8	2-3/8	3,000 - 3,500	7-5/8	17
4000	4-1/2 - 5-3/8	2-7/8	6,000 - 7,000	8-5/8	26
5000	5-1/2 - 7-3/8	3-1/2	7,000 - 9,000	9	40
7000	7-1/2 - 8-3/4	4-1/2	12,000 - 16,000	11-1/2	76
10000	10-1/4 - 12-1/4	6-5/8	28,000 - 32,000	16-1/2	125

Dimensions in inches, weights in lb (approximate). Column order follows the source layout.

DESIGN FEATURES

- **Tungsten carbide dressing:** for packers, retainers, bridge plugs and cement.
- **Standard API pin connection:** made up with a standard bit breaker; no extra sub needed.
- **Circulation:** flow along each blade and through the head centre for cooling and cuttings removal.
- **Stabilizing ribs:** above the blades, to protect the casing.
- **Cast construction:** not to be used as a junk mill; cast material has lower strength.

PERFORMANCE PARAMETERS

Weight on mill	2,000 - 8,000 lb
Practice	Light weight, low speed
Surface speed	250 - 340 ft/min
Rotary speed	Surf. speed / (O.D. x 3.14)
If bouncing	Cut speed, then raise slowly
Stabilizer	Above mill, O.D. not larger

HOW TO ORDER

Please specify: 1. mill dressed diameter 2. size and weight of the casing to run through, if known 3. protective subs, available on request.

JUNK MILL

CONVENTIONAL JUNK MILL

MILLING

GENERAL DESCRIPTION

THE WORKHORSE OF DOWNHOLE MILLING, USED TO MILL ALMOST ANYTHING IN THE HOLE INCLUDING CEMENT AND RUBBER PRODUCTS. IT IS THE ONLY TOOL ABLE TO MILL DRILL PIPE CEMENTED INSIDE AND OUT. IT IS ALSO USED TO MILL CAST-IRON BRIDGE PLUGS, AND TO MILL CASING THAT HAS BEEN MILLED TO THE POINT OF ROTATING, RUN ON A JOINT OF SLIGHTLY BENT DRILL PIPE. THREE DRESSING OPTIONS ARE LISTED: CONVENTIONAL, CONEBUSTER AND SUPER.

TECHNICAL CHARACTERISTICS

Cutting diameter (in.)	Top pin conn. (API Reg.)	Overall length: conventional and Conebuster (in.)	Overall length: Super (in.)	Fishing neck length (in.)	Fishing neck dia. (in.)	Wt (lb)
3-1/2 - 4-1/2	2-3/8	20	20	12	3	45
4-1/2 - 5-1/2	2-7/8	21	21	12	3-3/4	62
5-1/2 - 5-5/8	3-1/2	23	21	12	4-1/4	95
5-3/4 - 7-1/2	3-1/2	23	21	12	4-3/4	105
7-1/2 - 9	4-1/2	27	27	12	5-3/4	180
9-1/2 - 12-1/4	6-5/8	29	29	12	7-3/4	350
13 - 15	6-5/8 or 7-5/8	30	30	12	7-3/4 or 9-1/2	500
17 - 17-1/2	6-5/8 or 7-5/8	33	33	12	7-3/4 or 9-1/2	625
18-1/2 - 26	6-5/8 or 7-5/8	37	37	12	7-3/4 or 9-1/2	1,200

Dimensions in inches, weights in lb (approximate).

DESIGN FEATURES

- **Fishing neck:** allows retrieval of the mill if it is left in the hole.
- **Dressing options:** conventional, Conebuster or Super.
- **Inside casing:** mill head O.D. equal to the casing drift; wear pads of the same O.D. protect the casing; stabilizer of the same O.D. run directly above.
- **Bridge plugs:** mill O.D. about 1/8 in. under the plug size, to avoid skinning the casing.

OPERATING NOTES

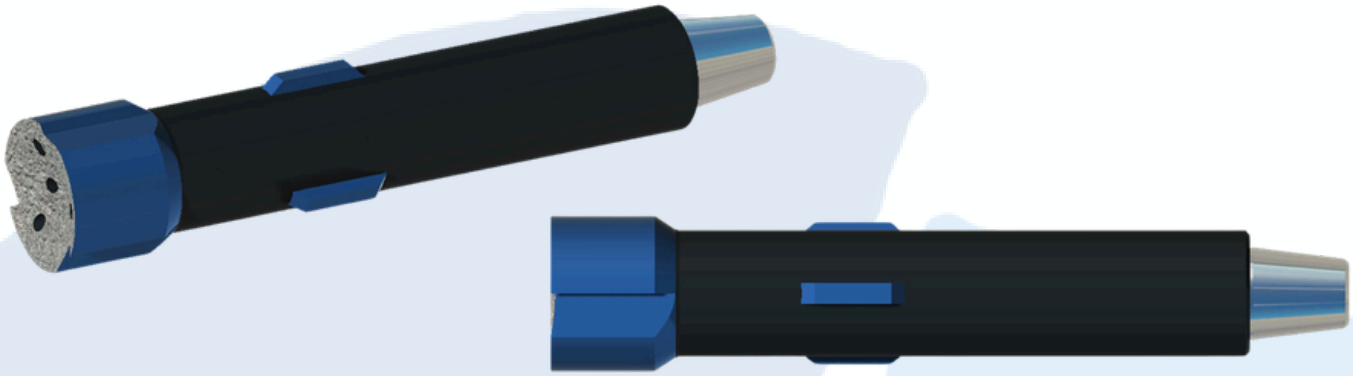
- **Open hole:** mill O.D. about 1/8 in. under hole diameter; at least 10,000 lb of drill collars; junk sub directly above the mill.
- **Spudding:** frequent spudding pounds loose junk onto bottom for effective milling; a rise in torque shows junk alongside the mill.
- **Starting:** set down on the junk while turning, then bring weight up; never apply weight before rotating.

PERFORMANCE PARAMETERS

Weight on mill	4,000 - 10,000 lb
Start-up rotary speed	60 - 80 rpm
Rotary speed once milling	80 - 100 rpm, normal pump
Weight, second pass	4,000 - 6,000 lb
Surface speed	250 - 340 ft/min
Rotary speed	Surf. speed / (O.D. x 3.14)
Loose junk cycle	Mill 1-2 ft, lift 15-20 ft, settle
Stationary junk cycle	Each 3-5 ft, lift 10-15 ft, ream
Rough running	Cut speed/weight, raise slowly
Stabilizer spacing	60 - 90 ft above mill (vertical)

HOW TO ORDER

Please specify: 1. mill diameter 2. size and weight of the casing to run through, if available 3. top connection 4. dressing (conventional, Conebuster or Super).



SUPER JUNK MILL

MILLING

The Junk Mill is a high-performance milling tool designed to grind and remove metal debris and obstructions in the wellbore, such as stuck tools, casing fragments, or drill bits. It is used in tough environments where standard mills might not suffice, offering superior cutting power and durability.

Dressed OD	Top Pin Connection
3½" - 4 ¼"	2⅜" API REG
4½" - 5½"	2⅜" API REG
5½" - 7½"	3½" API RG
7½" - 9"	4½" API REG
9½" - 12¼"	6⅝" API REG
13" - 15"	6⅝", 7⅝"
17" - 17½"	6⅝", 7⅝"
18½" – 26"	6⅝", 7⅝"



Conventional



Conebuster



Super

USAGE RECOMMENDATIONS

The Junk Mill should be used with a robust circulation system to clear away milling debris efficiently.

It is often recommended to combine the tool with a **Fishing Magnet** or **Junk Basket** to collect any metallic debris left behind.

Stabilizers are also useful for ensuring tool stability during operation, allowing for more consistent cutting action and prolonging the life of the mill.

TYPES OF DEBRIS MILLED

- Broken drill bits
- Casing fragments
- Stuck tools
- Fishing equipment debris
- Miscellaneous metallic obstructions

VIPER PILOT MILL

CASING REMOVAL AND SLOT RECOVERY MILL

MILLING

GENERAL DESCRIPTION

MILLING TOOL DESIGNED FOR THE EFFICIENT REMOVAL OF DOWNHOLE CASING STRINGS. IT IS USED MAINLY FOR OFFSHORE PLATFORM SLOT RECOVERY, WHERE INTERMEDIATE CASING IS REMOVED BACK TO THE SURFACE STRING OR CONDUCTOR SO THE WELL CAN BE RE-DRILLED TO A NEW TARGET INSTEAD OF BEING ABANDONED. WHERE THE CASING CEMENT IS POOR A MIX OF RETRIEVAL AND MILLING IS USED; WHERE CASING IS CEMENTED BACK TO THE SPOOLS, MILLING ALONE IS USED. THE BLADES CARRY A BRICK-PATTERN CARBIDE CUTTING STRUCTURE FOR HIGH PENETRATION RATE, IDEAL CUTTING SIZE AND LONG MILLING LIFE.

TECHNICAL CHARACTERISTICS

Tool series	Casing size	Body dia.	Blade dia.	No. of blades	Top pin conn.	Fishing neck length	Fishing neck dia.	Overall length	Wt (lb)
4500	4-1/2	4-3/4	5.250	3	3-1/2 IF BU	12	4-3/4	36	100
5000	5	4-3/4	5.813	3	3-1/2 IF BU	12	4-3/4	36	110
5500	5-1/2	4-3/4	6.300	3	3-1/2 IF BU	12	4-3/4	36	120
6000	6	4-3/4	6.875	3	3-1/2 IF BU	12	4-3/4	42	150
6600	6-5/8	5-3/4	7.640	3	4-1/2 IF BU	12	5-3/4	42	175
7000	7	5-3/4	7.906	3	4-1/2 Reg. BU	12	5-3/4	42	190
7600	7-5/8	6-1/4	8.750	5	4-1/2 IF BU	18	6-1/4	48	250
8600	8-5/8	6-3/4	9.875	5	5-1/2 Reg. BU	18	6-3/4	48	275
9600	9-5/8	8	10.875	5	6-5/8 Reg. BU	18	8	60	300
10700	10-3/4	9	12.000	5	7-5/8 Reg. BU	18	9	60	325
11700	11-3/4	10-1/4	13.000	5	8-5/8 Reg. BU	18	10-1/4	60	375
13300	13-3/8	11-1/2	14.625	5	8-5/8 Reg. BU	18	11-1/2	72	400
16000	16	14	17.250	5	8-5/8 Reg. BU	18	14	72	425

Dimensions in inches, weights in lb (approximate). Pilot stabilisation and guide or taper mill are dressed to the casing drift diameter. stabiliser sleeve available on casing sizes 7-5/8 in. and larger.



DESIGN FEATURES

- **Cutting structure:** carbide inserts in a brick pattern, developed to resist early wear and breakage.
- **Blades:** high-grade alloy steel, cutting edge set at the angle for maximum efficiency.
- **Extended blade length:** more footage per mill.
- **Stabiliser:** sleeve-type, changeable at the rig site; useful on eccentric casing strings.
- **Deviated holes:** blade design does not skin the next casing string.

OPERATING PARAMETERS

Weight on mill	5,000 - 10,000 lb
Cutting surface speed	250 - 340 ft/min
Rotary speed	From surface speed and casing O.D.

MILL STABILITY

- **Taper mill:** run directly ahead of the mill to clear junk or enlarge the casing stub.
- **Stabiliser:** same diameter as the mill, run immediately above it.
- **Sizing:** O.D. and stabilisation diameter set to avoid damage to outer casing strings.

HOW TO ORDER

Please specify: 1. tool series 2. size and weight of the casing to be milled 3. top pin connection.

PILOT MILL

PILOT MILL FOR TUBULARS AND LINERS

MILLING

GENERAL DESCRIPTION

MILLING TOOL WITH A GUIDE NOSE (PILOT) THAT ENTERS THE TUBULAR BEING MILLED AND KEEPS THE BLADES CENTRED ON IT. IT IS USED TO MILL WASHPIPE, SAFETY JOINTS, CROSSOVER SWAGES, WASHOVER SHOES, ADAPTERS, CASING, LINERS, DRILL PIPE AND SWAGED CASING. LINER HANGERS CAN BE MILLED EFFICIENTLY, AVOIDING INSIDE CUTS AND SPEAR RUNS. THE NOSE CAN BE DRESSED TO MILL ANY JUNK MET ALONG THE WAY.

TECHNICAL CHARACTERISTICS

Blade O.D.	Top pin conn. (API reg.)	Pilot O.D.	Overall length	Fishing neck length	Wt (lb)
3-1/4 - 3-7/8	2-3/8	1-3/4 - 2-3/4	27	12	40
4-1/8 - 4-3/8	2-3/8	1-3/4 - 2-3/4	27	12	45
4-1/8 - 5-3/8	2-7/8	2-1/8 - 3-1/4	27	12	120
5-1/2 - 5-5/8	3-1/2	2-1/2 - 4-3/4	38	16	240
5-3/4 - 7-3/8	3-1/2	2-1/2 - 4-3/4	38	16	255
6-1/8 - 9-7/8	4-1/2	4-3/4 - 6-3/4	42	18	305

Dimensions in inches, weights in lb (approximate). Standard API regular pin. Other sizes on request.

DESIGN FEATURES

- **Guide nose:** the pilot sits in the tubular bore and centres the blades.
- **Blade size:** about 1/4 in. over the O.D. of the tool joint or coupling to be milled.
- **Pilot size:** same as the drift diameter of the tubular.
- **Dressable nose:** can be dressed to mill junk.
- **Fishing neck:** standard neck for handling and recovery.
- **Typical jobs:** washpipe (most efficient tool for stuck washpipe), liners and hangers, adapters, swaged casing, drill collars with open I.D.

HOW TO ORDER

Please specify: 1. size and weight of the casing to be milled 2. size and weight of the casing to run through, if known 3. top pin connection.

OPERATING PARAMETERS

Rotary speed (start)	80 - 100 rpm
Tool weight (start)	2,000 - 6,000 lb or less
H-40 / K-55 casing	4,000 - 6,000 lb; 80 - 100 rpm
N-80 / P-110 / Q-135 casing	8,000 - 10,000 lb; 100 - 120 rpm
Entering the tubular	about 30 rpm, 2,000 lb
Typical milling rate (casing)	8 - 12 ft/hr

OPERATING NOTES

- **Alignment check:** rotate at about 125 rpm above the tubular to find neutral torque, then enter the pilot slowly; a gradual slow-down when rotation stops shows good alignment.
- **Steady feed:** apply weight at a constant rate, do not drill off.
- **Cuttings:** below a shoe, ream up and down every 15 - 20 ft milled; also in holes deviated 30 degrees or more.
- **Rate drop:** may be a loose steel ring turning with the mill; spud gently. If cutting stops with no torque rise, a tubular section may be turning: pull and retrieve with a spear.
- **Wear pattern:** hook wear means efficient cutting; taper wear means ease off the weight.

TAPER MILL

CP AND CT TAPER MILLS

MILLING

GENERAL DESCRIPTION

TAPER MILLS REMOVE RESTRICTIONS AND MILL THROUGH PINCHED OR COLLAPSED CASING. THEY CARRY A TAPERED OR SHORT BLUNT NOSE THAT SERVES AS A GUIDE. TWO VERSIONS: THE CP, WITH A SHORT BLUNT NOSE AND A SHORT TAPER, PRODUCES MUCH LESS TORQUE AND CAN TAKE MORE WEIGHT, SUITING THE TOUGHEST JOBS WHERE A LONG NOSE MIGHT BREAK; THE CT, WITH SPIRAL BLADES AND A POINTED NOSE, IS DESIGNED TO MILL THROUGH RESTRICTIONS.

TECHNICAL CHARACTERISTICS

Dressed O.D.	Top pin conn. (API reg.)	Length CT	Length CP	Fishing neck length	Fishing neck O.D.	Wt CT (lb)	Wt CP (lb)
3-1/4 - 3-7/8	2-3/8	34	30	10	3	80	60
4 - 4-3/8	2-3/8	34	30	10	3-1/8	90	70
4-1/2 - 5-3/8	2-7/8	38	31	10	3-3/4	106	75
5-1/2 - 5-5/8	3-1/2	42	32	13	4-1/4	155	115
5-3/4 - 6-3/8	3-1/2	44	32	13	4-3/4	160	120
6-1/2 - 7-3/8	3-1/2	46	34	13	4-3/4	170	130
7-1/2 - 7-7/8	4-1/2	54	36	13	5-3/4	250	185
8-3/8 - 9-1/2	4-1/2	54	36	14	5-3/4	280	220
9-3/8 - 9-7/8	4-1/2 or 6-5/8	54	36	14	5-3/4 or 7-3/4	345	280
10-3/8 - 11	6-5/8	57	38	14	7-3/4	415	355
11-1/2 - 12-1/4	6-5/8	60	40	14	7-3/4	455	390
14-3/4 - 15	6-5/8	70	60	18	7-3/4	525	460
17 - 17-1/2	6-5/8	70	60	18	7-3/4	595	530
20 - 26	6-5/8 or 7-5/8	76	66	18	7-3/4 or 9-1/2	1,250	1,125

Dimensions in inches (overall lengths in inches), weights in lb (approximate). Ranges as dressed diameters of each size group.

DESIGN FEATURES

- **CP:** blunt nose, short taper; low torque, tolerates more weight.
- **CT:** spiral blades and pointed nose for restrictions.
- **Typical jobs:** reaming collapsed casing and liners, cleaning permanent whipstock windows, milling jagged or split shoes, enlarging restrictions through retainers and adapters.
- **Stabilising:** keep 1/2 to 1 in. under drift diameter.

OPERATING PARAMETERS

Weight range (general)	2,000 - 4,000 lb
Start rotation above collapse	75 rpm (min. 75 rpm)
Weight increase in collapse	1,000 - 2,000 lb, slowly
Collapsed casing	about 50 rpm; 2,000 - 3,000 lb
Whipstock window clean-up	40 rpm; under 1,000 lb
Retainers / adapters	about 70 rpm; 2,000 - 6,000 lb
After clearing restriction	80 - 100 rpm

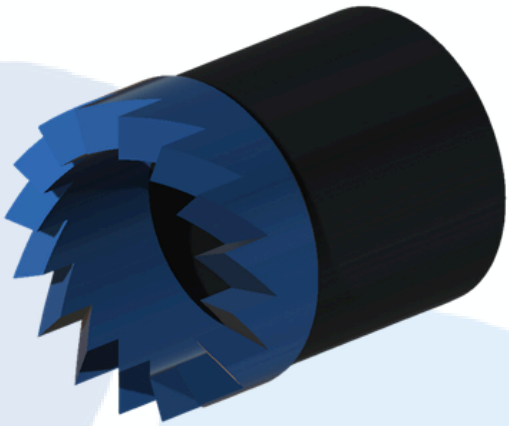
OPERATING NOTES

Run a mill about 1/4 in. larger than the minimum I.D. of the collapse and enlarge in stages; do not use a taper mill if the collapse has passed centre. A string taper mill can be used if there is a danger of sidetracking. Governing factor is torque: reduce speed and weight if it rises. For a window, run the same diameter as the largest window mill, rotate slowly its full length and repeat until torque is minimal; do not try to make hole.

HOW TO ORDER

Please specify: 1. type (CP or CT) 2. dressed diameter 3. top pin connection 4. fishing neck diameter, if different from standard.



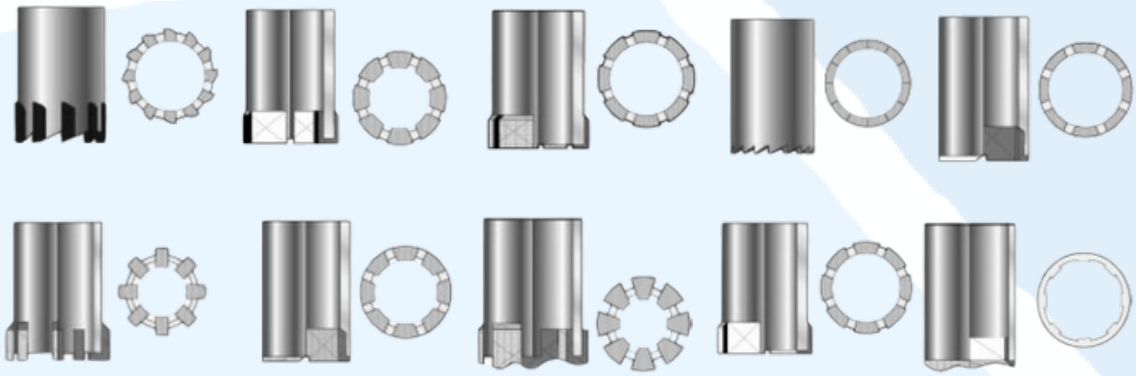


WASHOVER SHOE

MILLING

The Washover Shoes are designed in various sizes and styles to meet the many conditions encountered in oil well fishing and washover operations. Located at the bottom of the washover string, Washover Shoe is used to cut a clearance between the fish and the wall of the well bore during retrieval operations or to release stuck downhole objects. Washover Shoe can be dressed with carbide Inserts for hard milling and for best work duration.

Washover Extensions, Washover Pipes, and Top Subs are available according to the connection size of the washover string desired. Appropriate washover lifting equipment are also available.



MAIN APPLICATIONS

- Recovering stuck drill pipe, tubing, or casing.
- Clearing wellbore restrictions caused by collapsed formations or debris.
- Assisting fishing operations by providing a clean passage around stuck equipment.

TECHNICAL FEATURES

- Cutting edge designed for efficient material removal.
- Available in multiple sizes to match casing and tubing diameters.
- Hard-faced or carbide-enhanced surfaces for durability and cutting efficiency.
- Optimized fluid ports to maintain circulation and aid in debris removal.
- Compatible with various thread connections for seamless integration into the fishing string.

Fishing Magnet

PERMANENT MAGNET FOR JUNK RETRIEVAL

FISHING

GENERAL DESCRIPTION

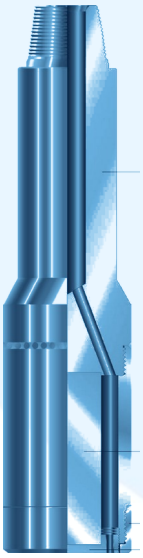
PERMANENT-MAGNET TOOL THAT RETRIEVES UNDRILLABLE OBJECTS WITH MAGNETIC ATTRACTION: BIT CONES, BEARINGS, SLIPS, MILL CUTTINGS, TONG PINS AND HAMMERS. IT PICKS UP SMALL, ODD-SHAPED ITEMS THAT INSIDE OR OUTSIDE CATCH TOOLS CANNOT HOLD. AVAILABLE FROM 1 IN. TO 20 IN. O.D., ON PIPE OR WIRELINE, WITH PULLS FROM 5 LB TO 5,000 LB DEPENDING ON SIZE. IT IS ALSO USED TO CLEAR THE HOLE BEFORE DIAMOND CORING.

TECHNICAL CHARACTERISTICS

O.D.	Hole size	Top pin	Approx. pull (lb)	Wt (lb)
1	1-1/4 to 2	5/8 11-N.C.	5 - 7	1
1-1/4	1-5/8 to 2-1/2	5/8 S.R.	8 - 10	1-1/2
1-1/2	2 to 2-3/4	5/8 S.R.	11 - 14	3-1/4
1-3/4	2-3/8 to 3-1/4	3/4 S.R.	15 - 20	5
2-1/4	2-3/4 to 3-5/8	3/4 S.R.	25 - 50	16
2-1/2	3 to 3-3/4	3/4 S.R.	50 - 85	18
3	3-1/4 to 4-1/8	2-3/8 Tbg	85 - 190	20
3-1/4	4 to 4-1/4	2-3/8 Tbg	85 - 190	25
3-1/2	4-1/4 to 4-1/2	2-3/8 API Reg.	150 - 250	27-1/2
4	4-1/2 to 5	2-3/8 API Reg.	175 - 250	43
4-1/2	5-1/8 to 5-1/2	2-7/8 API Reg.	250 - 320	67
5	5-5/8 to 6	2-7/8 API Reg.	320 - 385	80

O.D.	Hole size	Top pin	Approx. pull (lb)	Wt (lb)
5-1/2	6-1/8 to 6-1/2	3-1/2 API Reg.	385 - 425	95
5-3/4	6-1/8 to 6-1/2	3-1/2 API Reg.	385 - 425	106
6	6-5/8 to 7-1/2	3-1/2 API Reg.	425 - 500	120
7	7-5/8 to 8-1/2	4-1/2 API Reg.	550 - 700	162
8	8-5/8 to 9-3/4	4-1/2 API Reg.	700 - 850	205
9	9-7/8 to 11-5/8	4-1/2 API Reg.	850 - 1,000	335
10	10-1/2 to 11-7/8	6-5/8 API Reg.	1,000 - 1,125	390
10-1/2	11-3/4 to 13	6-5/8 API Reg.	1,125 - 1,260	431
11-1/2	12-1/4 to 14	6-5/8 API Reg.	1,260 - 1,550	512
14	15	6-5/8 API Reg.	1,700 - 2,100	770
16	17	6-5/8 API Reg.	3,200 - 4,000	1,200
19	20	13-3/8 V-4 thd	4,200 - 5,000	1,700

Dimensions in inches, weights in lb (complete assembly, approximate). S.R.: sucker rod pin; Tbg: tubing pin. Pulls assume the magnet face fully covers a flat surface; pick-up capacity drops considerably on smaller contact areas.



DESIGN FEATURES

- **Focused field:** flux is concentrated at the bottom pole plate; the outer case is not magnetised, so the tool works inside cased hole.
- **Circulation:** ports through the body exit inside the guide and keep the pole plate clean.
- **Guides:** flush (standard) for large or flat objects; lipped for 3 or more cones or leaning objects; mill guide for tong dies, 1 - 2 cones and safety runs before diamond coring.
- **Run on pipe or wireline:** wireline needs adaptors and gives no circulation.

OPERATING PARAMETERS

Approach	Stop 6 in. to 1 ft above the fish; circulate to wash
Contact	Reduce flow, lower onto fish, rotate lightly, stop pumps
Max. weight, rotating	5,000 lb
Max. weight, no rotation	10,000 lb
Proper charge	2,500 gauss or more

Excess weight or rotation can damage the pole plate and magnetic element. Never store two magnets face to face: the smaller loses charge. Clean ports after each run.

HOW TO ORDER

Please specify: 1. tool name and assembly number 2. O.D. 3. top connection if other than standard. Recommended accessories: 1 lipped guide and 1 mill guide.

STRING MAGNET

IN-LINE STRING MAGNET

CLEANING

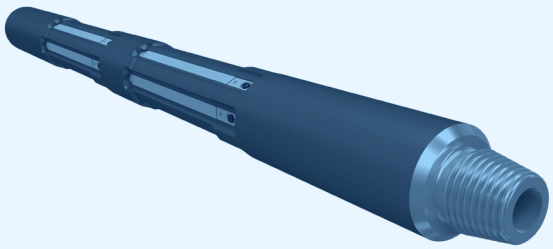
GENERAL DESCRIPTION

IN-LINE MAGNET RUN IN THE WORK STRING TO COLLECT FERROUS DEBRIS DURING CASING CUTTING, MILLING, PLUG AND ABANDONMENT AND CASING EXIT OPERATIONS, ESPECIALLY WHERE THE AVAILABLE CIRCULATION IS NOT ENOUGH TO CARRY THE SWARF OUT OF THE HOLE. LARGE MAGNETIC SLEEVES WITH RARE EARTH MAGNETS GIVE A HIGH PICK-UP CAPACITY. THE BODY IS A SINGLE PIECE WITH NO INTERNAL CONNECTION AND IS RATED FOR HIGH TORQUE. AVAILABLE FOR CASING FROM 4-1/2 TO 14 IN., IN STANDARD OR HIGH TEMPERATURE VERSIONS.

TECHNICAL CHARACTERISTICS

Casing size (in)	Tool OD (in)	Connection	Tool yield torque (ft-lb)	Tool tensile (lb)	Connection yield torque (ft-lb)	Connection tensile* (lb)
4-1/2, 4-3/4, 5	3-1/8	2-3/8 API Reg	15,000	457,940	5,810	362,660
5-1/2, 6	3-3/4	2-7/8 API Reg	20,000	698,880	11,080	478,540
6-5/8, 7	4-3/4	3-1/2 API IF	25,000	1,032,640	18,090	572,690
7-5/8	5	3-1/2 API IF	42,500	2,467,780	18,090	572,690
9-5/8	6-1/2	4-1/2 API IF	80,000	2,126,520	38,060	919,880
10-3/4	8	6-5/8 API Reg	105,000	3,759,840	83,380	1,770,500
13-3/8, 13-5/8, 14	10-1/8	6-5/8 API Reg	250,000	4,879,680	83,380	1,770,500

* Tensile load at zero make-up torque. Dimensions in inches, loads in lb and ft-lb.



DESIGN FEATURES

- **Large magnetic area:** high pick-up capacity for ferrous debris.
- **Rare earth magnets:** powerful field, standard and high temperature options.
- **Replaceable sleeves:** magnetic sleeves are easily removed and replaced.
- **One-piece body:** solid mandrel with no internal connection.
- **High torque rating:** suited to milling and cutting strings.
- **Fishable through the ID.**

APPLICATIONS

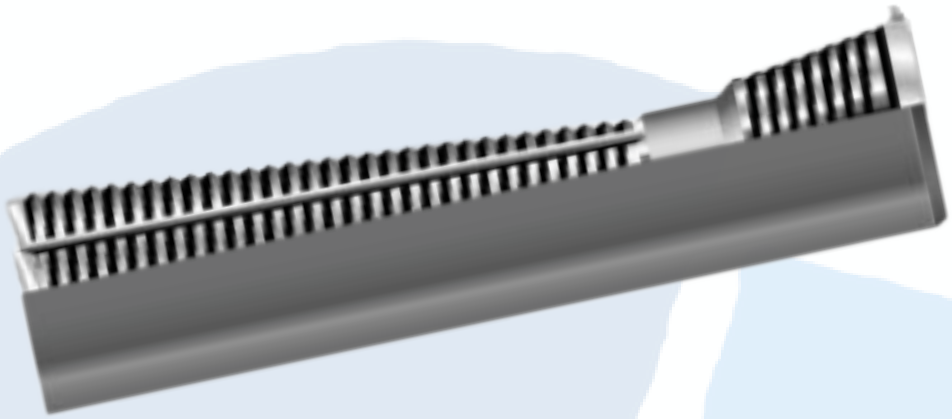
- Casing cutting and milling
- Plug and abandonment
- Casing exit (window milling)
- Wellbore cleanout where circulation is limited

BENEFITS

- Better ferrous debris removal when circulation is inadequate
- Cleaner wellbore, less non-productive time
- Lower overall cost of the operation

HOW TO ORDER

Please specify: 1. casing size and weight 2. tool OD and top / bottom connections 3. standard or high temperature magnets 4. number of magnetic sleeves and spare sleeves.



DIE COLLAR

RECOVERY

GENERAL DESCRIPTION

The Die Collar is an external catch tool designed to engage the O.D. of a fish. It cuts a thread into the fish to assure proper engagement.
Die Collars can be designed with a threaded bottom end for setting a guide or an oversize guide to assure fish engagement when the hole size is considerably larger than the diameter of the fish.

SPECIFICATIONS

Top Connection	2-3/8 REG	2-7/8 REG	3-1/2 IF
Fishing Neck OD	3-1/8	3-3/4	4-3/4
Max. ID Wickers	2-5/8	3-1/4	4-1/8
Mini. ID Wickers	1-1/4	1-1/2	2-1/2
Top Connection	4-1/2 IF	6-5/8 REG	7-5/8 REG
Fishing Neck OD	6-1/2	8	9-1/2 - 11
Max. ID Wickers	5-1/2	7	8-1/4 - 9-3/4
Mini. ID Wickers	3-1/4	4-1/2	5-3/4 - 7

USAGE RECOMMENDATIONS

Any item that can be retrieved by external engagement can be engaged by the Die Collar. The fish must have a clean O.D., long enough to allow entry of the fish into the Die Collar.
In operation it is necessary only to run the Die Collar in the hole to the top of the fish, apply less than one point of weight and rotate sufficiently to embed the tapered threads of the collar onto the fish, cease rotation and pull the fish.



HOOK

RECOVERY

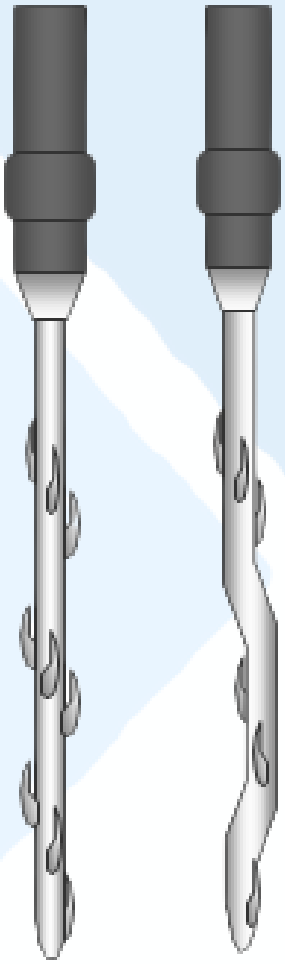
GENERAL DESCRIPTION

This tool is used for retrieving wire line. It is designed with barbs welded onto the straight or crankshaft shank.

Hooks are available for running through various casing sizes and with various barbs sizes in order to catch all types of wire lines.

SPECIFICATIONS

Into Casing	5 Csg	7 Csg	9-5/8 Csg
Overall OD	4-1/4	6-1/8	8-1/4
Hook Max Catch	1-1/2	3/4	1-1/2
Connection	2-7/8 IF	3-1/2 IF	4-1/2 IF



Overshot

SERIES 150 RELEASING AND CIRCULATING OVERSHOT

FISHING

GENERAL DESCRIPTION

EXTERNAL-CATCH FISHING TOOL THAT ENGAGES, PACKS OFF AND PULLS TWISTED-OFF OR LOST TUBING, DRILL PIPE, COUPLINGS, TOOL JOINTS, CASING AND OTHER SMOOTH-O.D. FISH. A SPIRAL OR BASKET GRAPPLE GRIPS THE FISH OVER A LONG SECTION, SO HEAVY PULLING, TORSION AND JARRING LOADS ARE TAKEN WITHOUT DAMAGING THE TOOL OR THE FISH. ENGAGING AND RELEASING BOTH USE RIGHT-HAND ROTATION ONLY.

FULL STRENGTH SIZES

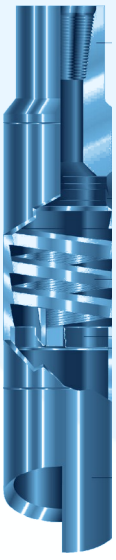
Overshot O.D.	Max. catch, spiral	Max. catch, basket	Type	Wt (lb)*
3-5/8	2-1/2	2	FS	55
4-1/8	3-1/8	2-5/8	FS	70
5-9/16	4-1/4	3-11/16	FS	130
5-5/8	4-5/8	4-1/8	FS	130
5-3/4	4-3/4	4-1/4	FS	138
7-7/8	6-1/4	5-1/2	FS	261
8-1/8	6-1/2	5-3/4	FS	274
8-1/4	6-5/8	5-7/8	FS	280
9-5/8	8	7-1/4	FS	337
10-1/8	8-1/2	7-3/4	FS	358
10-5/8	9	8-1/4	FS	401
11-1/4	9-5/8	8-7/8	FS	520
11-3/4	10-1/8	9-3/8	FS	642

* Assembly dressed with spiral parts. Inches. FS = full strength. Slim-hole and semi full strength types also exist; undersize parts fit smaller fish.

WIDE CATCH VERSION

Tool O.D.	Catch range	Max. catch, spiral	Max. catch, basket	Load, spiral (lb)
4-11/16	1/4	3-21/32	3-1/16	332,100
5-3/4	5/16	4-3/4	4-1/8	407,100
7-3/8	3/16	6-1/2	5-7/8	435,200
8-1/8	5/16	7	6-1/4	548,900
9-1/8	5/16	8	7-1/4	548,900
11-3/4	1/2	10-1/8	9	969,000

Same design with an extended catch range, up to four times that of a standard overshot, to reach heavily worn fish. Standard or high-pressure seals, full flow through the fish, coarse top-sub threads with set screws, 18 in. fish neck. Can cut grapple inventory by up to 75%.



DESIGN FEATURES

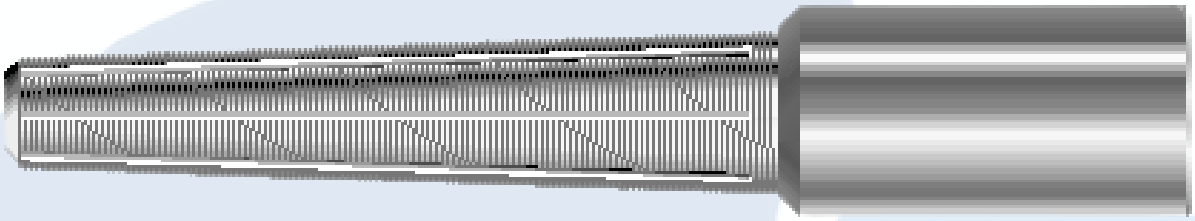
- **Spiral or basket grapple:** spiral for fish near maximum catch size, basket for smaller fish.
- **Pack-off:** sleeve packer with spiral grapple; double-lip mill control packer (field-replaceable seals) with basket grapple.
- **Circulation** through the fish to help free it.
- **Right-hand only:** no left-hand rotation to engage or release.
- **Accessories:** oversize, wall hook and mill guides, extension subs, lock rings; coarse-thread version for easier assembly.

OPERATION

- **Engage:** rotate slowly right while lowering over the fish, let the twist slack out, then pull.
- **If the fish stays:** start the pumps and keep an upward strain.
- **Release:** bump down, then rotate right while slowly raising.
- **Precautions:** stop the pumps before lowering over the fish; bump full string weight before releasing; no left-hand rotation without upward strain.

HOW TO ORDER

Please specify: 1. tool name and assembly number 2. size and type of fish 3. top connection 4. O.D. if other than standard. Spares (spiral): 3 packers, 2 grapples per size, 1 control. Spares (basket): 2 grapples, mill control packers per size.



TAPER TAP

RECOVERY

GENERAL DESCRIPTION

The Taper Tap is an internal catch fishing tool designed to retrieve tubular members from the well bore.

It is the most simple, rugged, dependable and economical internal catch fishing tool for freeing fish.

Skirt type Taper Tap is threaded for a skirt. A skirt is used when the hole size is drastically different from the fish size. The skirt is assembled to the Taper Tap and will assist in guiding the fish for proper engagement and retrieval.

USE

Assemble the Taper Tap to the bottom of the fishing string. Run in the hole until reaching the fish.
Apply minimum weight (less than one point) and rotate sufficiently to allow the wicker threads to be embedded into the interior surface of the fish. Once the fish has been securely engaged, it can be lifted from the hole.

SPECIFICATIONS

Normal size	5/8 SR	2-7/8” REG	3-1/2” REG	4-1/2” REG	4-1/2” IF	6-5/8” FH
OD	1-1/2”	3-3/4”	4-1/4”	5-1/2”	6-1/8”	8”
Total Length	16"	27"	33"	36"	36"	38"
Smallest Wicker	3/8"	7/8"	1"	1-3/4"	3-1/4"	4-5/8"
Larger Wicker	1-1/4"	1-7/8"	2-1/2"	3-3/8"	4-3/4"	6-3/8"

Spear

FULL CIRCLE RELEASING SPEAR

FISHING

GENERAL DESCRIPTION

INTERNAL-CATCH FISHING SPEAR FOR TUBING, DRILL PIPE AND CASING OF ALL SIZES. FULL CIRCLE SLIPS AND THE TAPERS ON THE BODY SPREAD THE EXPANSION STRAIN OVER A LONG VERTICAL SECTION AND ALMOST THE WHOLE INNER CIRCUMFERENCE OF THE FISH, ALLOWING HEAVY PULLING AND JARRING WITHOUT DISTORTING IT. CAN BE RUN WITH INTERNAL CUTTERS FOR CUT-AND-PULL OPERATIONS.

DRILL PIPE AND TUBING

Spear size	Slips Ø retracted	Slip area (sq in)	Wt (lb)	Body yield (lb)
2-3/8 DP	1-11/16	55	32	56,400
2-3/8 Tbg	1-29/32	59	40	71,500
2-7/8 DP	2	74	55	110,400
2-7/8 Tbg	2-5/16	87	60	134,000
3-1/2 DP	2-7/16	95	71	172,000
3-1/2 Tbg	2-13/16	115	92	224,000
4 DP	3-3/32	126	107	296,000
4 Tbg	3-5/16	135	127	275,000
4-1/2 DP	3-15/32	178	180	318,000
5-1/2 DP	4-1/2	265	270	628,000

CASING

Spear size	Slips Ø retracted	Slip area (sq in)	Wt (lb)	Body yield (lb)
4-1/2	3-25/32	210	132	376,000
5	4	217	169	454,000
5-1/2	4-1/2	265	226	628,000
6	-	315	283	945,000
6-5/8	5-1/2	350	358	1,060,000
7	5-3/4	365	398	1,228,000
7-5/8	6-7/16	430	502	1,414,000
8-5/8	7-1/4	485	652	1,761,000
9-5/8	8-3/16	580	833	2,324,000
10-3/4	9-1/2	690	1,240	3,300,000
11-3/4	10-9/16	785	1,552	4,300,000
13-3/8	12	890	2,017	5,961,000

Dimensions in inches. DP: drill pipe, Tbg: tubing. Yield strengths are theoretical body values (accuracy about 20%), not the strength of the fish; jarring may multiply the pull load by 3 to 10. Each spear has 3 to 6 slips depending on size.



DESIGN FEATURES

- **Full circle slips:** large engagement area, low risk of rupturing the fish.
- **Friction grip:** friction springs up to 3-1/2 in. nominal and 4 in. tubing; friction blocks on larger sizes and casing.
- **Safety ring:** stops the spear entering a fish too small to release from.
- **Two release modes:** bump down and rotate; left-hand wickers on the slips as back-up.
- **Circulation:** safety ring packoff seals inside the fish.

ACCESSORIES

Mill, side hill and sub-type nuts; 36 in. extensions (bored for circulation); skirts and guides for oversize holes; stop subs.

OPERATION

- **Engage:** lower slowly into the fish, rotate at least 1/6 turn to the left, then pull.
- **Release:** bump down full string weight, rotate at least 1/6 turn to the right and raise.
- **Deep or crooked holes:** more rotation may be needed; it does no harm.
- **Always** keep a heavy upward strain when rotating the fish, and bump down before releasing.

HOW TO ORDER

Please specify: 1. spear name and assembly number 2. size and type of top connection 3. size and weight of the pipe to be caught. Recommended spares: 2 sets of slips per fish size, 1 unitizer (or 2 friction block assemblies), 2 friction retainers, 1 stop, 1 safety ring per slip size.



T-TANG

RECOVERY

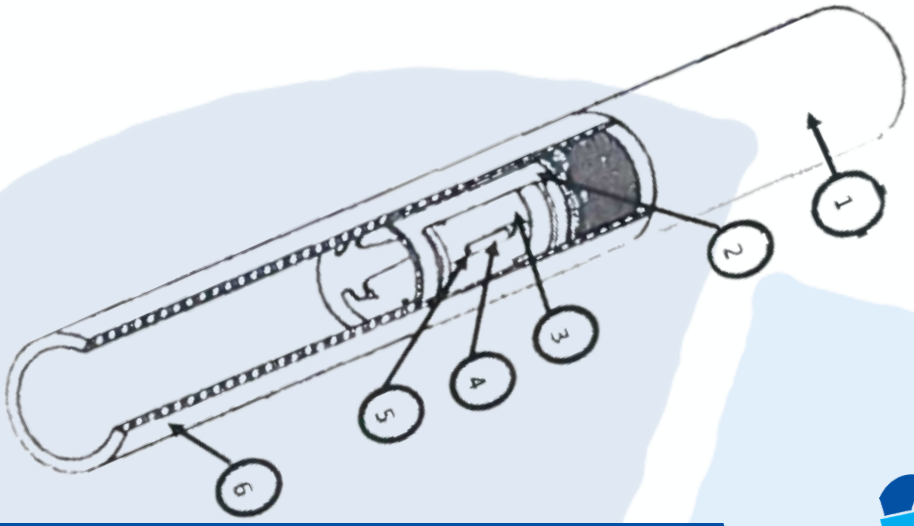
GENERAL DESCRIPTION

The T-Tang (also known as T-Catch in the fishing tools industry) is a fishing tool designed to engage and lock into a T-shaped groove machined onto the object to be retrieved downhole.

Unlike the Overshot, which envelops and grips the fish from the outside via a grapple, or the Spear, which inserts inside the fish, the T-Tang works through direct mechanical engagement: its T-shaped tang locates into the matching profile machined onto the piece to be fished. Its use therefore requires that the downhole component was designed with this T-profile in mind (a sub, mandrel, or specific downhole equipment).

SPECIFICATIONS

Model	OD	Conn up (upper sub)	Conn down (upper sub)	Conn up (lower sub)	Conn down (lower sub)
1	4-3/4"	4-1/2" FJWP Box	4-3/4" FJWP Pin	4-3/4" FJWP Box	4-1/2" FJWP Pin
2	5-3/4"	5-1/2" FJWP Box	5-1/2" FJWP Pin	5-1/2" FJWP Box	5-1/2" FJWP Pin



T-DOG 4-11/16

RECOVERY

GENERAL DESCRIPTION

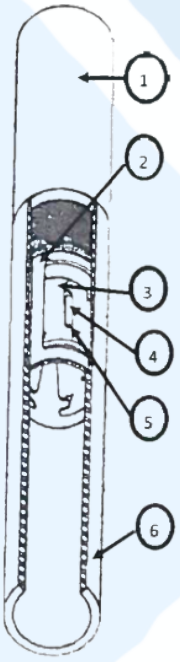
Bull Dog's T-Dog Overshots are design to assist the operator in catching Pipe or Tubing. Easy to operate and to redress in the field.

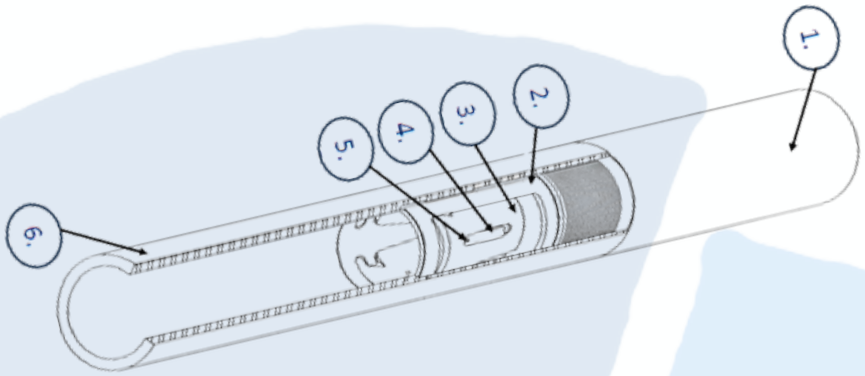
Bull Dog's T-Dog Overshots are design to catch collared pipe or tubing and release the fish if necessary. The operation maybe repeated several without risk of preventing the T-Dog from re-engaging the fish.

- Heat treated alloy Steel.
- Flame harden slips
- Box and Pin connection for easy placement in the workstring.

SPECIFICATIONS

N°	Description
1	4-11/16 T-Dog Top Sub with 4-1/2" Wash Pipe Connection
2	2-7/8 T-Dog Cage
3	2-7/8 T-Dog
4	4-11/16 T-Dog Screws
5	4-11/16 T-Dog Bowl 4-1/2 WP Connection
6	4-11/16 T-Dog Assembly





T-DOG 5-3/4

RECOVERY

GENERAL DESCRIPTION

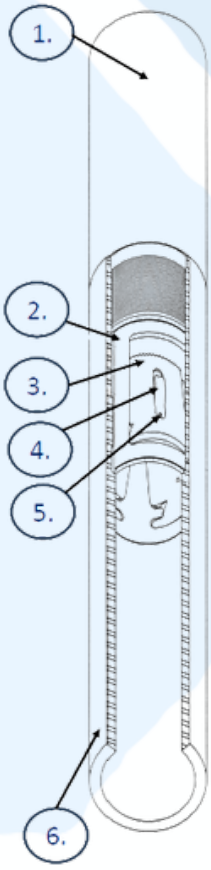
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Bull Dog's T-Dog Overshots are design to catch collared pipe or tubing and release the fish if necessary. The operation maybe repeated several without risk of preventing the T-Dog from re-engaging the fish.

- Heat treated alloy Steel.
- Flame harden slips
- Box and Pin connection for easy placement in the workstring.

SPECIFICATIONS

N°	Description
1	5-3/4 T-Dog Assembly (Bowl & Top with 5-3/4 WP, BxP)
2	T-Dog Cage 2-3/8 - 2-7/8
3	T-Dog 2-3/8 - 2-7/8
4	T-Dog Spring
5	T-Dog Screws
6	T-Dog Bowl - 5-3/4 Wash Pipe



PDM

POSITIVE DISPLACEMENT MOTOR WITH SLEEVE STABILIZER

Enedril

DRILLING

GENERAL DESCRIPTION

POSITIVE DISPLACEMENT MUD MOTOR (PDM) FOR DIRECTIONAL AND PERFORMANCE DRILLING, FROM 3-1/8 TO 9-5/8 IN. IN STOCK. THE MOTOR IS BUILT FROM A POWER SECTION (ROTOR AND STATOR), AN ADJUSTABLE OR FIXED BENT HOUSING, A DRIVESHAFT ASSEMBLY, A SEALED OR MUD-LUBRICATED BEARING ASSEMBLY AND THE BIT BOX. THE BEARING HOUSING CARRIES A THREAD FOR A SCREW-ON SLEEVE STABILIZER (STRAIGHT OR SPIRAL BLADE) OR OFFSET PAD, USED TO TUNE THE BUILD RATE AND TO REDUCE LATERAL VIBRATION. SPEED IS PROPORTIONAL TO FLOW, TORQUE TO DIFFERENTIAL PRESSURE.

TECHNICAL CHARACTERISTICS

Power section OD (in)	Lobes	Stages	Max. flow (gpm)	Max. bit speed (rpm)	Std. max diff. press. (psi)	Std. torque (lb-ft)	High-power diff. press. (psi)	High-power torque (lb-ft)
3-3/8	7/8	3.0	120	188	460	782	690	1,170
3-3/4	6/7	3.0	200	160	440	1,314	660	1,970
4-3/4	7/8	2.6	300	79	390	3,500	590	5,250
5	6/7	8.0	350	302	1,215	3,500	1,820	5,250
6-1/4	7/8	2.9	500	82	425	6,277	640	9,420
6-1/2	8/9	4.0	500	145	605	4,674	910	7,010
6-3/4	7/8	5.7	600	150	845	8,204	1,270	12,310
7	7/8	7.5	600	186	1,120	8,508	1,680	12,760
7-3/4	7/8	4.0	900	149	570	8,333	860	12,500
8	6/7	5.0	900	159	715	9,618	1,070	14,430
9-5/8	6/7	6.0	1,200	155	880	15,775	1,320	23,660
11-1/4	6/7	5.6	1,800	162	820	19,377	1,230	29,070

One configuration per size: the highest-torque standard power section listed. Other lobe and stage combinations are available. High-power = enhanced elastomer power section (about 50% higher differential pressure). Values per manufacturer motor tables, for reference.

DESIGN FEATURES

- **Modular motor:** power section, bent housing, driveshaft, bearing assembly and bit box can be selected to suit the job.
- **Adjustable bend:** field-adjustable housing (2.38, 3 or 4 degrees) for a wide range of build rates; fixed housings on request.
- **Sleeve stabilizer option:** threaded bearing housing takes a screw-on straight or spiral blade sleeve, or an offset pad.
- **Bearing assembly:** oil sealed (100% of the flow to the bit) or mud lubricated.
- **Driveshaft:** conventional or high-torque version for the strongest power sections.
- **Dump sub and rotor catch:** bypass while tripping; retrieval of the motor after a housing failure.

PERFORMANCE PARAMETERS

Speed	Proportional to flow rate
Torque	Proportional to differential pressure, independent of flow
More torque	More stages, enhanced elastomer or even-wall stator
Stall	Standpipe pressure rises, penetration stops
If stalled	Stop rotation, slow or stop pumps, release torque slowly, lift off bottom

HOW TO ORDER

Please specify: 1. motor OD and power section (lobes, stages, standard or high-power) 2. bearing type 3. housing angle 4. sleeve stabilizer type and size, or offset pad 5. top sub / dump sub and bit box connection 6. fluid, temperature and flow range.

BOOST MWD

MUD PULSE MWD SYSTEM

Enedril

DRILLING

GENERAL DESCRIPTION

BOOST IS A POSITIVE MUD PULSE MWD SYSTEM THAT SENDS SURVEYS, DIRECTIONAL TELEMETRY, GAMMA RAY AND OTHER SENSOR DATA TO SURFACE IN REAL TIME, USING PRESSURE SIGNALS IN THE WELLBORE. THE CORE SYSTEM INCLUDES PULSERS, ELECTRONICS, SENSORS AND BATTERIES, WITH MANY CONFIGURABLE OPTIONS. ADJUSTABLE MODULE PLACEMENT ALLOWS ALTERNATIVE SENSOR POSITIONS, AND DOWN-LINKING ALLOWS ADJUSTABLE LOGGING SEQUENCES.

TECHNICAL CHARACTERISTICS

Outside diameter	47.625 mm (1.875")
Length, 1 battery	5.28 m (17.3 ft)
Length, 2 batteries	6.94 m (22.8 ft)
Pressure rating	137,895 kPa (20,000 psi)
Operating temperature	-20 to +185 °C (-4 to 347 °F)
High-temp. version	200 °C available
Rotary speed	60 rpm recommended
Wireline	Retrievable, re-insertable
Min. I.D. to retrieve	57 mm (2.25")
Bend radius	20° / 10 m
Collar sizes	89 - 203 mm (3.5" - 8")
Flow rate	0.28 - 4.45 m³/min (75 - 1,200 gpm)

Data transmission	Positive mud pulse
Power supply	Lithium 28 VDC / 3 A
Battery capacity	400+ h (2 battery modules)
Flow switch	Vibration, in pulser driver electronics
Shock limit	500 g, 0.5 ms, half sine, all axes
Vibration limit	5 - 30 Hz @ 1" d.a.; 30 - 500 Hz, 20 g
LCM tolerance	40 - 50 ppb, any size, premixed
Pulse width	0.250 - 2.00 s
Transmission time	10 - 28 s
Internal memory	24 MB
Programmable sequences	4
Top or bottom mounted	Both positive pulse versions

d.a.: double amplitude. Standard tools are rated to 185 °C; high-temperature options are available.

SENSOR SPECIFICATIONS

Parameter	Range	Accuracy	Spread
Inclination	0 - 180°	± 0.10°	0.20°
Azimuth	0 - 360°	± 0.50°	0.50°
Tool face	0 - 360°	± 0.20°	0.40°
Dip angle	0 - ± 90°	± 0.2°	0.40°
Magnetic field	0 - 100,000 nT	± 125 nT	250 nT
Temperature	-35 - 200 °C	± 0.5 °C	1.00

TELEMETRY PARAMETERS

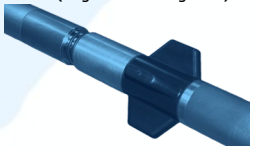
Inclination, azimuth, magnetic, dip angle and gravity qualifiers, raw X, Y, Z survey, toolface, continuous inclination and azimuth, battery status, shock, vibration, RPM, stick-slip, temperature, generic variables, annular pressure, orbital gamma (360° available).

DESIGN FEATURES

- **Field-proven design:** rugged and dependable, for aggressive drilling conditions.
- **Anti-plugging pulser:** motor driven, performs well in LCM and high-solids mud.
- **Simple operation:** wide range of flow rates and rotary speeds.
- **Ratcheting military connector:** robust barrel-to-barrel inter-module link, easy to assemble and service.
- **Stable in the hole:** precision-trimmed centralizers; shock mitigation absorbs vibration, reduces wear and improves MTBF.
- **Custom configuration:** adjustable module placement, down-linking, configurable options.

HOW TO ORDER

Please specify: 1. collar size 2. top or bottom mounted version 3. 1 or 2 battery modules 4. temperature rating (185 °C standard, 200 °C option) 5. sensor options (e.g. orbital gamma) and logging sequences.



WHIPSTOCK

RETRIEVABLE CASING SIDETRACK WHIPSTOCKS

Enedril

WHIPSTOCK

GENERAL DESCRIPTION

FAMILY OF RETRIEVABLE WHIPSTOCKS USED TO SIDETRACK OUT OF CASED HOLE, FOR RE-DRILLS, RE-ENTRIES AND MULTILATERAL WINDOWS. THE WHIPSTOCK IS RUN ON A STARTER MILL TOGETHER WITH ITS ANCHORING DEVICE, ORIENTED, SET, AND THE STARTER MILL IS SHEARED OFF THE WHIP; THE WINDOW IS THEN MILLED AND POLISHED ALONG THE WHIP FACE. THREE SETTING SYSTEMS ARE COVERED: A COMBINATION PACKER AND WHIPSTOCK (SET HYDRAULICALLY, SEALS THE HOLE), AN ANCHOR AND WHIPSTOCK (SET HYDRAULICALLY, NO SEAL) AND A MECHANICALLY SET BOTTOM-TRIP WHIPSTOCK. ALL HAVE A 3 DEGREE FACE ANGLE FOR POSITIVE KICK-OFF AND CAN BE RETRIEVED IN ONE TRIP. WINDOWS CAN BE MILLED THROUGH TWO STRINGS OF CASING.

TECHNICAL CHARACTERISTICS - PACKER TYPE

Casing size	Overall length	Body Ø	Packer length	Whip face length	Face angle (°)	Wt (lb)	Bypass valve length	Bypass valve Ø	Valve wt (lb)
5-1/2	182	4-3/16	76	106	3	585	25	3-3/8	40
7-5/8	216	5-5/8	84	133	3	980	43	4-3/4	150
7-5/8	229	5-15/16	84	146	3	1,400	43	4-3/4	150
9-5/8	261	8-5/8	84	178	3	2,500	46	6-3/4	240
13-3/8	338	11-3/4	87	251	3	6,595	38	8-1/4	400

TECHNICAL CHARACTERISTICS - MECHANICAL BOTTOM-TRIP TYPE

Casing size	Overall length	Body Ø	Anchor length	Whip face length	Face angle (°)	Wt (lb)
5 - 5-1/2	142-3/8	4-5/16	35-1/8	74	3	570
7	172-1/8	5-7/16	38-1/8	95	3	875
8-5/8	189-1/2	7-7/16	43-1/2	124	3	1,175

Dimensions in inches, weights in lb (approximate). Packer type: bypass valve data listed separately. Bottom-trip type can also be set permanently; 7-5/8 and 9-5/8 sizes on request.

DESIGN FEATURES

- **One trip:** anchor or packer, starter mill and whip run together; retrieved in one trip.
- **Anchor or packer:** large slip area reduces casing stress; ratchet ring and anti-rotation keys hold the whip.
- **Shear bolt:** confirms the anchor is fully set before milling starts.
- **Bottom-trip type:** sets with 3,600 lb on a trigger rod; no hydraulic pressure; slips retract on retrieval.
- **Orientation:** muleshoe sub, gyro or MWD.

SETTING PARAMETERS

Hydraulic set (packer, anchor)	3,000 - 3,500 psi
Bottom-trip set, shear-down	15,000 - 20,000 lb
Set depth	2 - 6 ft above a casing collar
Hole angle above 4°	Set within 90° of high side
Window milling	Starter, window, watermelon mills

HOW TO ORDER

Please specify: 1. casing size and weight 2. setting system (packer, anchor or bottom-trip) 3. connections 4. orientation method.

VENTURI BASKET

VENTURI JUNK BASKET

FISHING

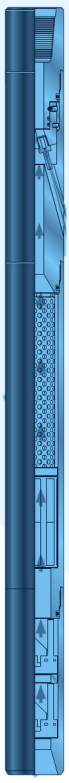
GENERAL DESCRIPTION

JUNK RETRIEVAL TOOL FOR COILED TUBING. FLUID PUMPED THROUGH THE STRING AND OUT OF NOZZLES IN THE VENTURI CHAMBER CREATES A VACUUM THAT DRAWS FLUID AND DEBRIS IN AT THE BOTTOM OF THE TOOL AND BACK UP THROUGH THE VENTURI TUBES; MOST OF IT MIXES WITH THE PRESSURIZED FLOW AND RECIRCULATES AROUND THE BOTTOM. IN EFFECT A HIGH-POWERED VACUUM CLEANER, IT WORKS WITH FLUID, MULTIPHASE FLUID, FOAM OR GAS. A HOLLOW MAGNETIC SECTION WITH A FINGER-TYPE TRAP HOLDS THE JUNK INSIDE THE TOOL, AND IT DOES NOT DEPEND ON HOLE SIZE: A 3-1/8 IN. OD TOOL CAN CLEAN UP 7 IN. CASING.

TECHNICAL CHARACTERISTICS

Size (OD)	ID	Top connection	Length	Tensile strength
1-3/4	0.469	1 AMMT box	33	36,000 lb
2-1/16	1	1-1/2 AMMT box	33	45,840 lb
2-1/4	1	1-1/2 AMMT box	33	45,840 lb
2-5/8	1	1-1/2 AMMT box	36-5/8	45,840 lb
3-1/8	1	2-3/8 PAC pin	42	109,600 lb
3-1/8	1	2-3/8 REG box	42	109,600 lb

Dimensions in inches. All sizes: operating pressure 5,000 psi, temperature rating 0 to 350 F, standard nozzle 0.062 in.



DESIGN FEATURES

- **Venturi suction:** nozzles create a vacuum that pulls fluid and debris into the tool from the bottom.
- **Interchangeable nozzles:** changed out to suit the available pump rate, fluid or gas.
- **Debris filter screen:** placed before the venturi chamber to keep debris out of the venturi tubes.
- **Magnetic trap:** hollow magnetic section with a finger-type trap retains junk; debris is carried out of the well inside the tool.
- **Barrel extensions:** extension tubes increase the volume of junk the tool can carry.
- **Hole size independent:** works in holes much larger than the tool OD.

HOW TO ORDER

Please specify: 1. tool size and top connection 2. nozzle size for the pump rate and fluid 3. extension tube length(s) for the junk volume to be carried 4. service kit.

OPTIONAL NOZZLES

Nozzle sizes (in.)	0.062 (std), 0.078, 0.094, 0.109, 0.125, 0.140, 0.156, 0.171, 0.187
Applies to	All sizes, 1-3/4 to 3-1/8 (two nozzle series: 1-3/4 to 2-1/4 and 2-5/8 to 3-1/8)

EXTENSION TUBES

Tool size	Extension tube lengths (ft)
1-3/4	4, 6
2-1/16	4, 6
2-1/4	4
2-5/8	2, 4, 5, 5
3-1/8	4, 10

OPERATING PARAMETERS

Fluids	Fluid, multiphase fluid, foam, gas
Pressure	5,000 psi
Temperature	0 to 350 F